MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SHIFT REGISTER, DIGITAL, KEY CODE PDM

THIS SPECIFICATION INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic shift register module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 6.43 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be PDM.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/28-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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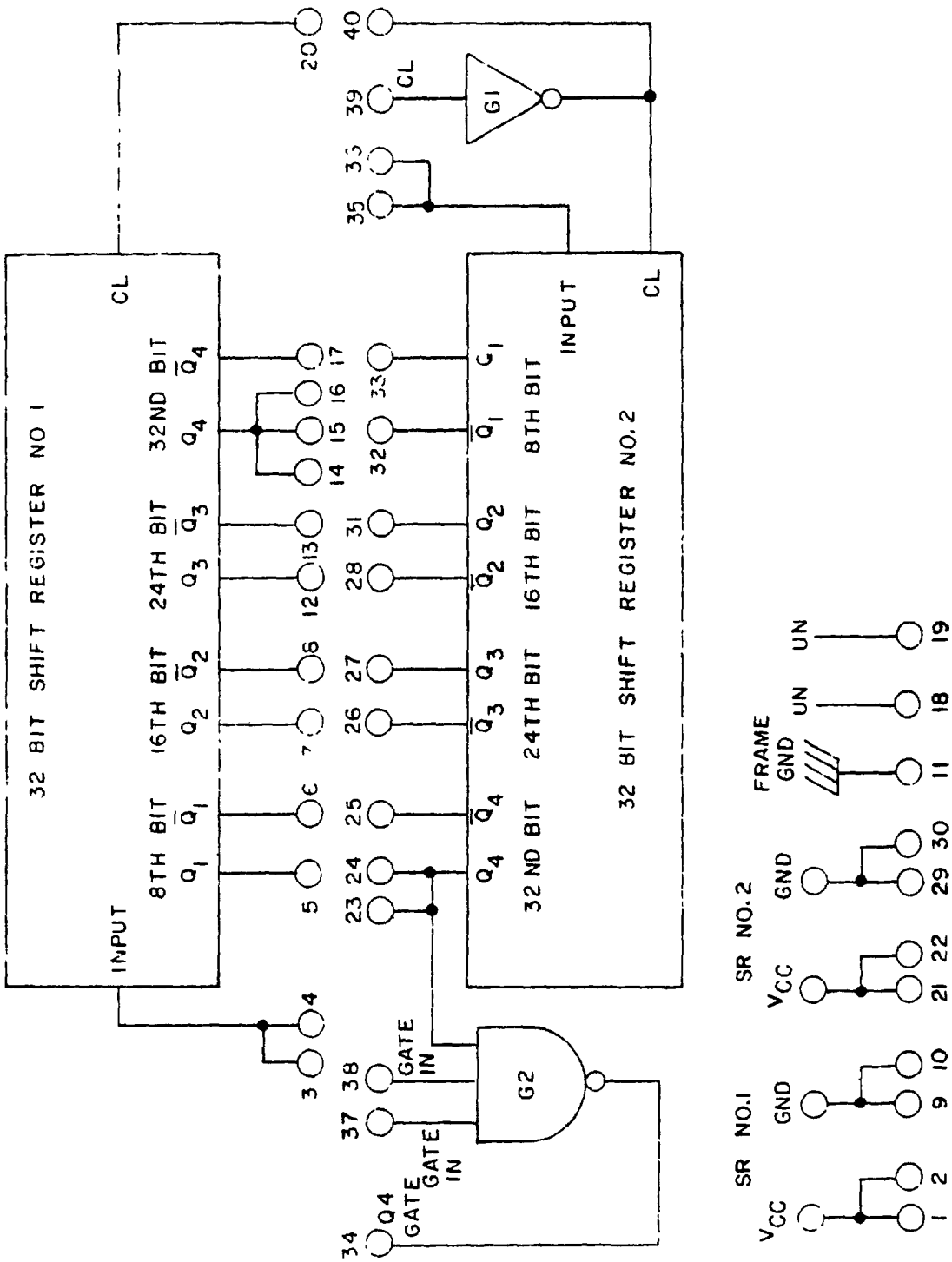


FIGURE 1 PDM module function diagram, shift register

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A16	Output voltage (low loaded)	III	S	0.1	0.8	0.1	0.8	0.1	0.8	0.1	0.8	V
B1-B5	Input current	IV	S	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	mA
B6	Input current	IV	S	0	2	0	2	0	2	0	2	mA
C1-C6	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D5	Input current, high level	VI	X	0	60	0	60	0	60	0	60	µA
D6	Input current, high level	VI	X	0	360	0	360	0	360	0	360	µA
E1, E2	Input current, low level	VI	X	0.10	1.73	0.10	1.73	0.10	1.73	0.10	1.73	mA
E3-E5	Input current, low level	VI	X	0.10	2.20	0.10	2.20	0.10	2.20	0.10	2.20	mA
E6	Input current, low level	VI	X	0.10	10.40	0.10	10.40	0.10	10.40	0.10	10.40	mA
F1-F16	Output short circuit current, high level	VI	X	15	80	15	80	15	80	15	80	mA
F17, F18	Output short circuit current, high level	VI	X	40	100	40	100	40	100	40	100	mA
G1-G8	Output voltage, high level (input threshold high)	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
G9-G20	Output voltage, high level (input threshold low)	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
H1-H8	Output voltage, low level (input threshold low)	VII	X	0.00	0.40	0.00	0.42	0.00	0.40	0.00	0.44	V
H9-H20	Output voltage, low level (input threshold high)	VII	X	0.00	0.40	0.00	0.42	0.00	0.40	0.00	0.44	V

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
J1	Power supply current, high level	VIII	X	100	345	100	345	100	345	100	345	mA
J2	Power supply current, low level	VIII	X	100	345	100	345	100	345	100	345	mA
K1-K10	Pin continuity	IX	X	0	1	0	1	0	1	0	1	
K11	Pin continuity	IX	X	0	1	0	1	0	1	0	1	
L	Isolation	IX	S	10	∞	1		10	∞	1	∞	MΩ
M1, M2	Transition time (t_{TLH})	X	S	2	20	2	20	2	25	2	25	ns
M3-M18	Transition time (t_{TLH})	X	S	5	20	5	20	5	25	5	25	ns
N1, N2	Transition time (t_{THL})	XI	S	2	20	2	20	2	25	2	25	ns
N3-N18	Transition time (t_{THL})	XI	S	5	20	5	20	5	25	5	25	ns
P1, P2	Propagation delay time (t_{PLH})	XII	S	2	25	2	25	2	30	2	30	ns
P3-P18	Propagation delay time (t_{PLH})	XII	S	5	57	5	57	5	62	5	62	ns
Q1, Q2	Propagation delay time (t_{PHL})	XIII	S	2	25	2	25	2	30	2	30	ns
Q3-Q18	Propagation delay time (t_{PHL})	XIII	S	5	73	5	73	5	78	5	78	ns
R1, R2	Maximum clock frequency and minimum set-up time check	XIV	S	25	98	25	98	25	103	25	103	ns
R3, R4	Maximum clock frequency and minimum set-up time check	XIV	S	25	121	25	121	25	126	25	126	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R5, R6	Maximum clock frequency and minimum hold time check	XIV	S	5	73	5	73	5	78	5	78	ns
R7, R8	Maximum clock frequency and minimum hold time check	XIV	S	7	98	7	98	7	103	7	103	ns
S	Power supply transient rise time	XV	S	2.5	7.5	2.5	7.5	2.5	7.5	2.5	7.5	µs
T1-T18	Output crosstalk, high level	XVI	Q	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
U1-U18	Output crosstalk, low level	XVI	Q	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
V	Power supply transient amplitude	XVII	Q	0	400	0	400	0	400	0	400	mV
W1-W7	Input capacitance	XVIII	Q	2	12	2	12	2	12	2	12	pF
W8	Input capacitance	XVIII	Q	2	48	2	48	2	48	2	48	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° +5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through K and M through U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through W shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5 PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6 NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.6).

6.3 Cross reference of classification. Military specification MIL-M-28787/28 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658885 dated 12 November 1971. Part number M28787/28-1 supersedes part numbers 2658885 and M28787/28. These parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits of table I.

Supply voltage	5 Vdc ±10%
Clock input	
Frequency	16 MHz maximum
Pulse width	35 ns minimum

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Input voltage	
Low level	0.8 V maximum
High level	2.0 V minimum
Input current	
Low level (V_{IL} = 0 V)	
SR inputs	1.72 mA maximum
Gate inputs	2.2 mA maximum
Clock input (ISR)	10.4 mA maximum
High level (V_{IH} = 4.5 V)	
SR inputs	60 µA maximum
Gate inputs	60 µA maximum
Clock input (ISR)	300 µA maximum
Output voltage	
Low level	0.4 V maximum
High level	2.4 V minimum, V_{CC} maximum
Q Output drive current	
Low level (V_{OL} = 0 V)	8.7 mA maximum
High level (V_{OH} = 2.4 V)	300 µA minimum
$\overline{Q}$ Output drive current	
Low level (V_{OL} = 0 V)	10.4 mA maximum
High level (V_{OH} = 2.4 V)	300 µA minimum
Gate output drive (Pin 34)	
Low level (V_{OL} = 0 V)	22 mA maximum
High level (V_{OH} = 2.4 V)	1.32 mA minimum
Power dissipation maximum	
Q outputs (high level)	1.9 W (outputs not loaded)
Q outputs (low level)	1.9 W (outputs not loaded)

6.5 Module functional requirements. This module consists of two independent 32-bit shift registers. True and complementary outputs are available at 8, 16, 24 and 32-bits for each register. Shift register No. 2 shall have a gated clock input (pin 39) and a gated 32-bit output (pin 34) with two inputs (pin 37 and 38). Shift register No. 1 shall not be gated. The internal module clock driver can be utilized in shift register No. 1 by externally tying pin 20 to pin 40. The clock pulse inverter and driver causes shift register No. 2 to shift information to the output on the negative edge of an input clock pulse at pin 39. A clock pulse at pin 20 causes shift register No. 1 to shift information to the output on the positive edge

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6.5.1 Input and output relationship. Each 32-bit shift register shall have the input and output relationship of table II for positive logic.

TABLE II. Input and output for positive logic.

Input	Output							
t_n	t_n +8		t_n +16		t_n +24		t_n +32	
	Q1	$\overline{Q1}$	Q2	$\overline{Q2}$	Q3	$\overline{Q3}$	Q4	$\overline{Q4}$
L	L	H	L	H	L	H	L	H
H	H	L	H	L	H	L	H	L

NOTES:

1. t_n = Bit time before clock pulse.
2. t_n + 8 = Bit time after 8 clock pulse.
3. t_n +16 = Bit time after 16 clock pulses.
4. t_n +24 = Bit time after 24 clock pulses
5. t_n +32 = Bit time after 32 clock pulses.
6. H = High level.
7. L = Low level.

6.5.2 Paralleling of outputs. Paralleling of outputs is not permissible

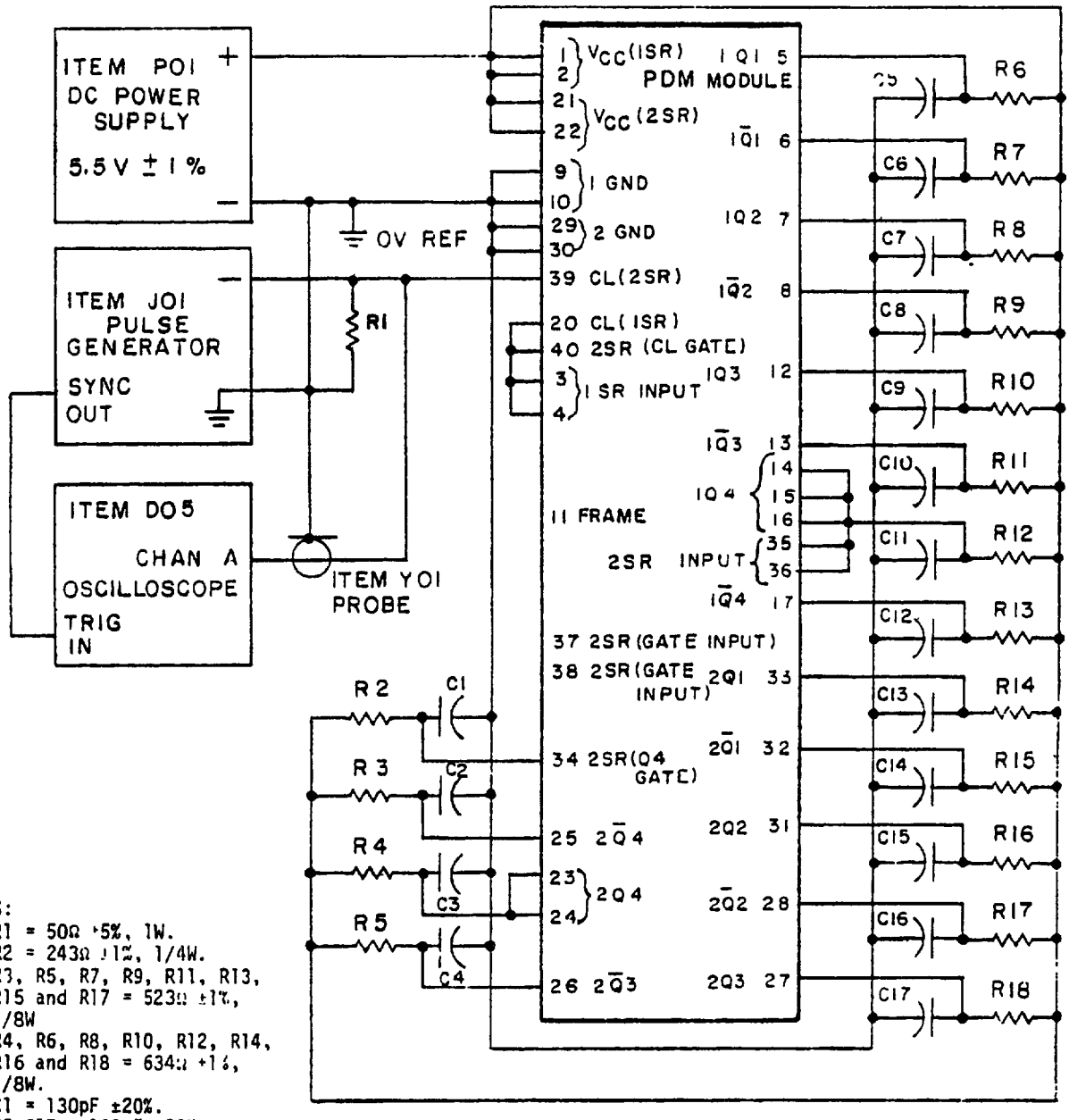
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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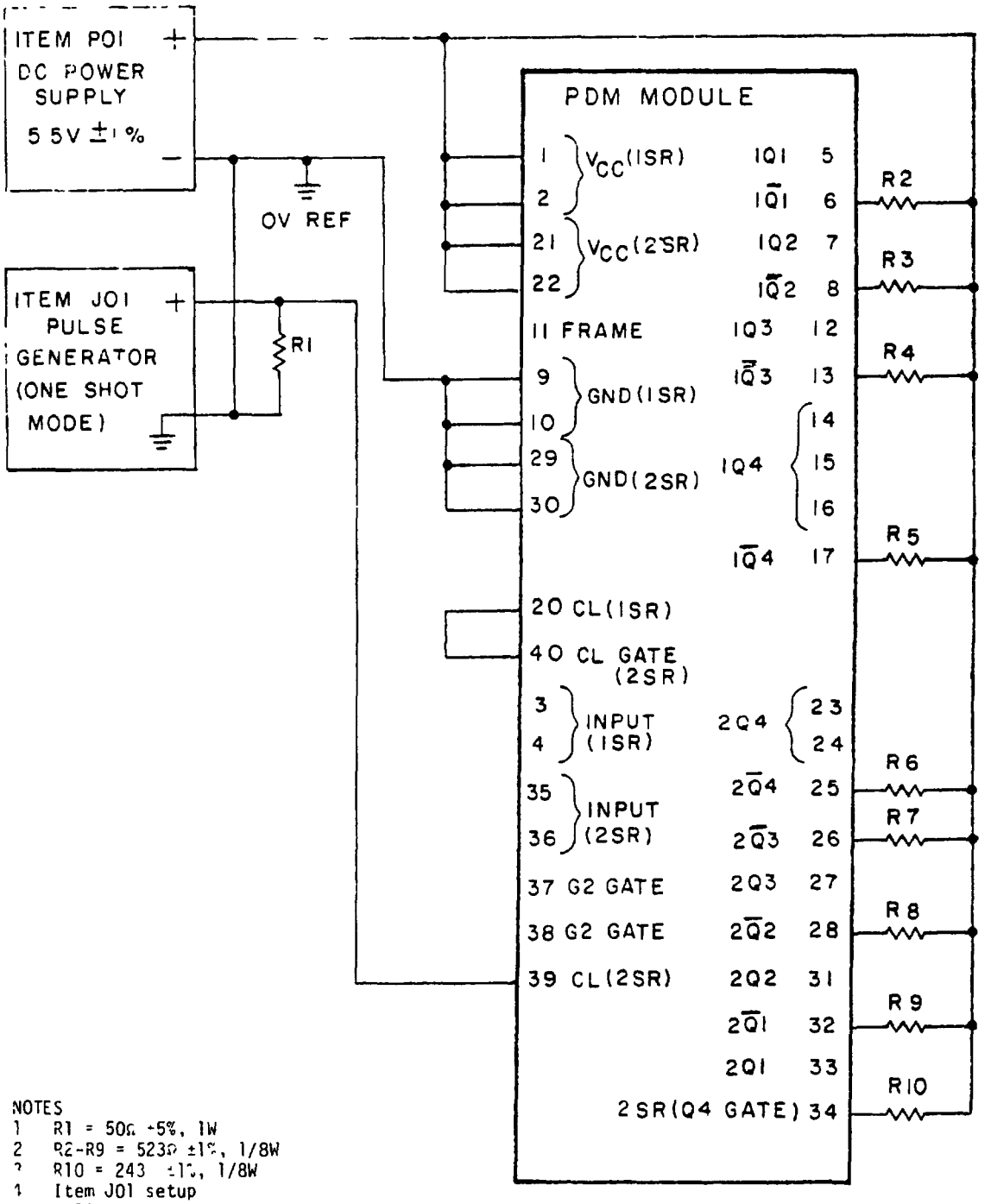
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NOTES:
1. R1 = 50Ω ±5%, 1W.
2. R2 = 243Ω ±1%, 1/4W.
3. R3, R5, R7, R9, R11, R13, R15 and R17 = 523Ω ±1%, 1/8W
4. R4, R6, R8, R10, R12, R14, R16 and R18 = 634Ω ±1%, 1/8W.
5. C1 = 130pF ±20%.
6. C2-C17 = 100pF ±20%.
7. Item J01 setup.
   PA = 4.0 ±0.5V
   PW = 2.5 ±0.5us.
   PRR = 200 ±20 kHz.
   $t_{TLH} = t_{THL} = 25 \pm 5$ns.
8. RC loads are required for operating temperature and life tests and not required for operating burn-in test

FIGURE 2. Operating temperature and life test circuit

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NOTES
1. R1 = 50Ω ±5%, 1W
2. R2-R9 = 523Ω ±1%, 1/8W
3. R10 = 243 ±1%, 1/8W
4. Item J01 setup
   PA = 4.0 ±0.5V
   PW = 1.0 ±0.1µs
   $t_{TLH} = t_{THL}$ = 25 ±5ns
5. Pulse pin 39 for a minimum of 32 clock pulses before measurements are taken (sets $\overline{Q}$ outputs at low level)

FIGURE 3 Component temperature test circuit

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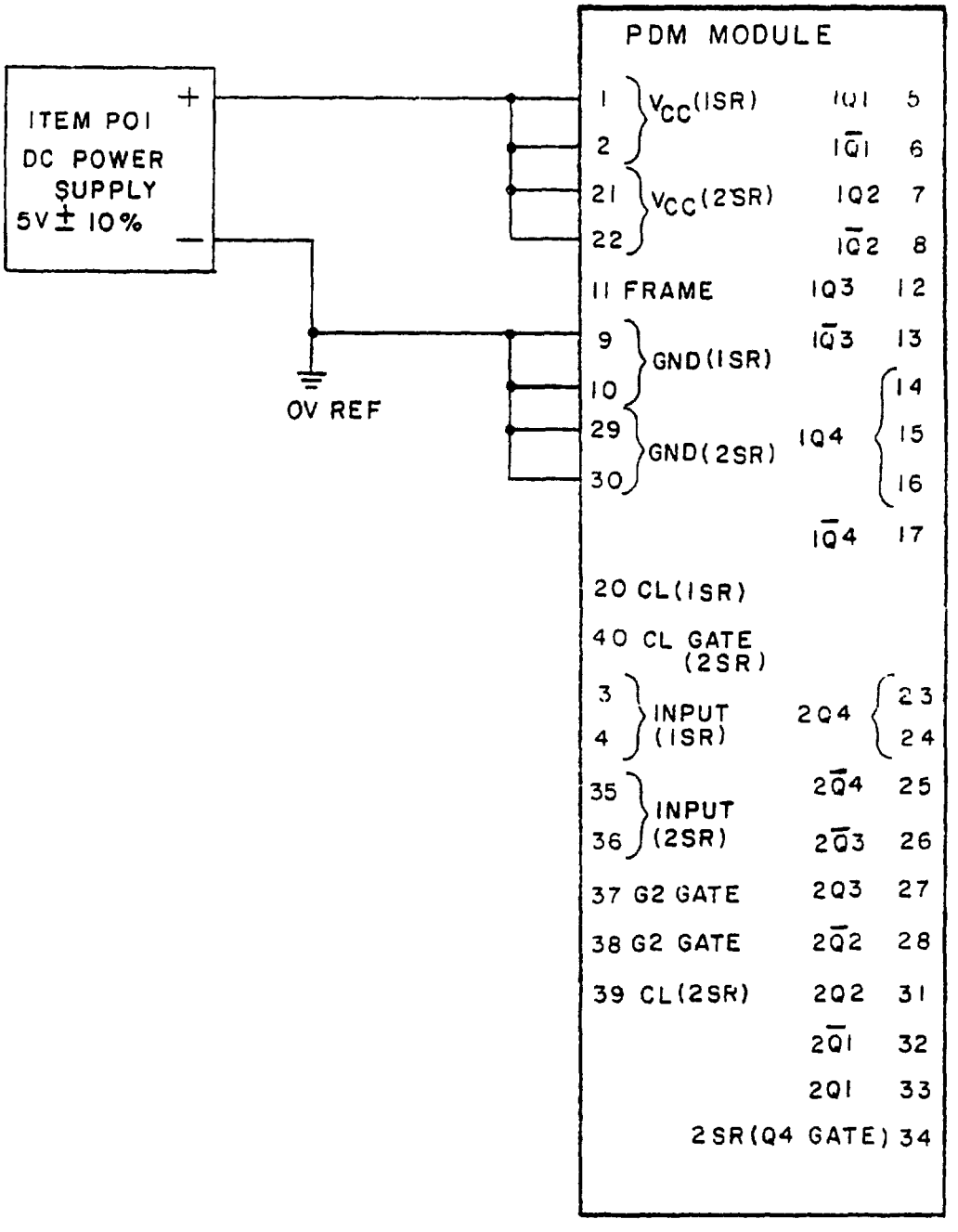


FIGURE 4 Vibration test circuit.

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TABLE III Output voltage (low loaded) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
A1	1Q1 Output voltage (low loaded)	3,4	6	Open	6	Open	Open	0.1	0.8	V	5
A2	1Q2		8		8						
A3	1Q3		13		13						
A4	1Q4		17		17						
A5	2Q1	35,36	32		32						
A6	2Q2		28		28						
A7	2Q3		26		26						
A8	2Q4		25		25						
A9	2SR (GATE Q4)	35,36, 37,38	34		Open	34					
A10	1Q1	Open	5	5		Open	3,4				
A11	1Q2		7	7			3,4				
A12	1Q3		12	12			3,4				
A13	1Q4		14,15, 16	Open	14,15,16		3,5				
A14	2Q1		33	33	Open		35,36				
A15	2Q2		31	31							
A16	2Q3		27	27							
A17	2Q4		23,24	23,24							
A18	2SR (GATED CL)	39	20,40	Open			Open				

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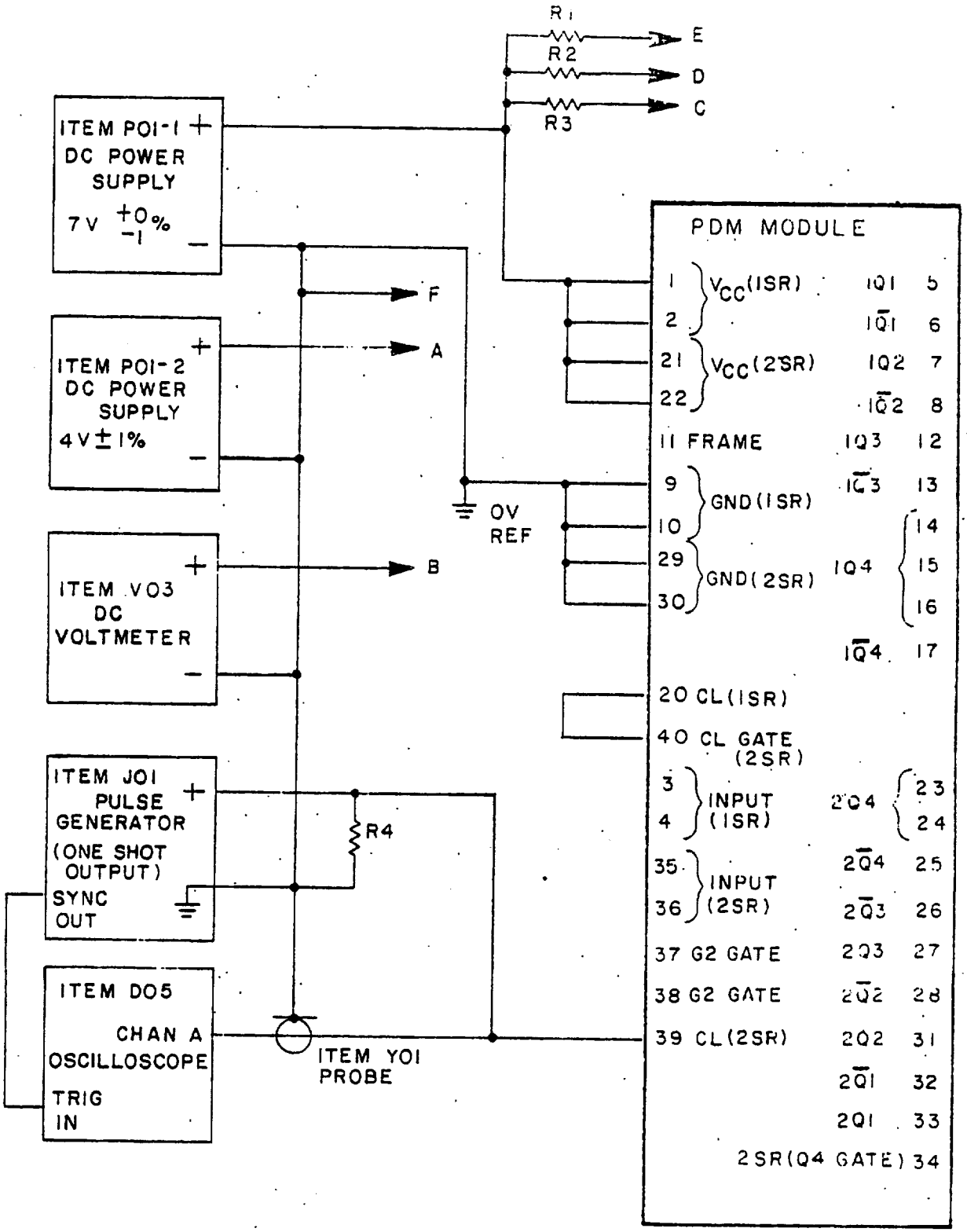


FIGURE 5. Output voltage (low loaded) test circuit for table III.

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NOTES
1 R1 = 243 Ω 1%, 1/4W
2 R2 = 523 Ω 1%, 1/8W
3. R3 = 634 Ω 1%, 1/8W
4 R4 = 50 Ω 5%, 1W
5 Item J01 setup
PA = 4 0 ±0.5V
PW = 1 0 ±0 1 s
$t_{TLH} = t_{THL} = 25 \pm 5$ns.
6 Before the measurement is recorded and after the module clock is pulsed (8 pulses for test conditions A1, A5, A10 and A14, 16 pulses for test conditions A2, A6, A11, and A15, 24 pulses for test conditions A3, A7, A12, and A16; 32 pulses for test conditions A4, A8, A13, and A17 and no pulses for test condition A18), the output shall be monitored for a minimum of 5 seconds

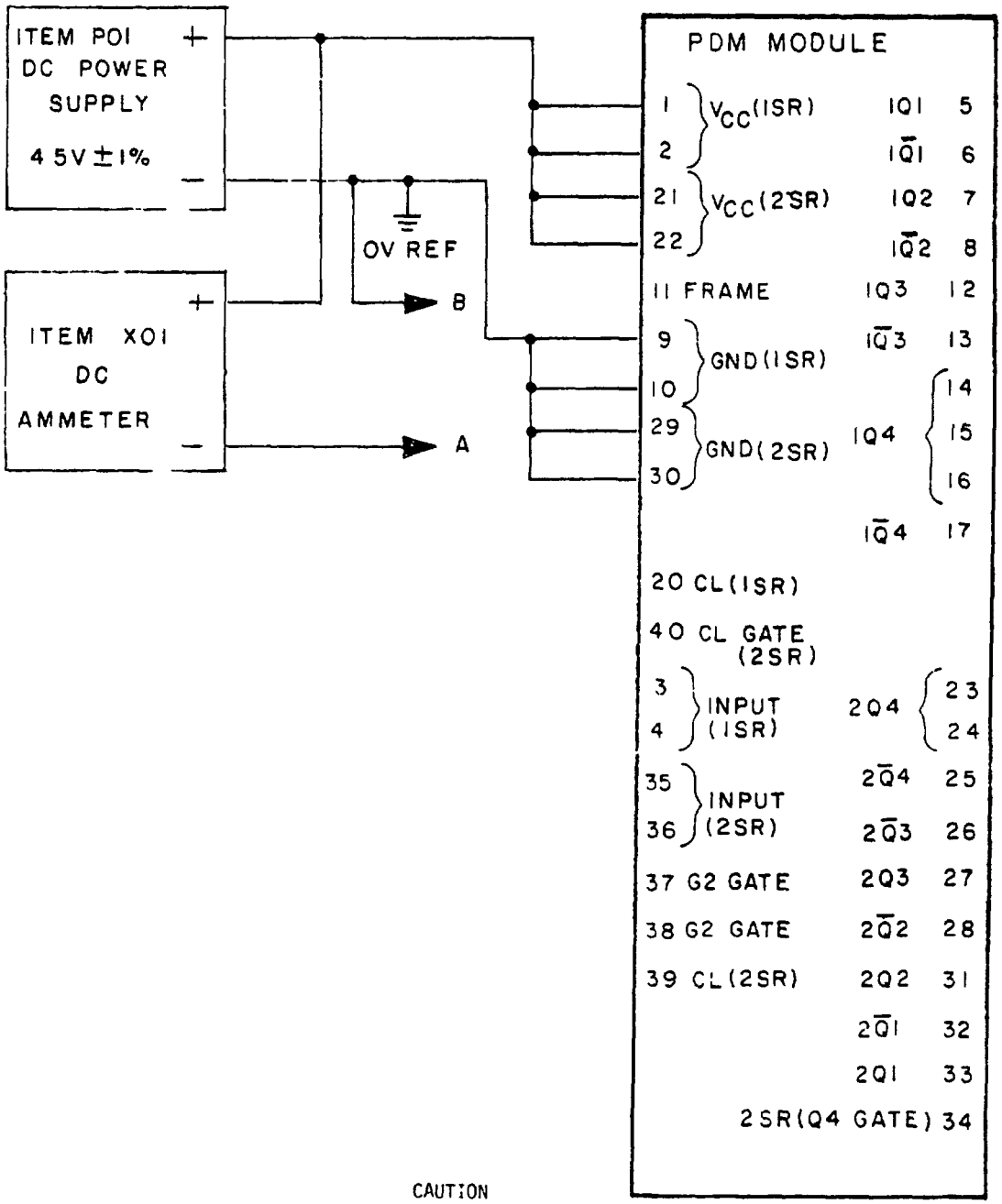
FIGURE 5 Output voltage (low loaded) test circuit for table III - Continued

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TABLE IV. Input current test.

Test condition	Parameter		Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
B1	1SR INPUT	Input current	3,4	20	0.0	0.5	mA	6
B2	2SR INPUT	↓	35,36	Open	↓	↓	↓	↓
B3	2SR GATE Q4	↓	37	38	↓	↓	↓	↓
B4	2SR GATE Q4	↓	38	37	↓	↓	↓	↓
B5	2SR CL	↓	39	Open	↓	↓	↓	↓
B6	1SR CL	↓	20	3,4	↓	2.0	↓	↓

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CAUTION

Do not isolate the chassis of ammeter X01 from earthground

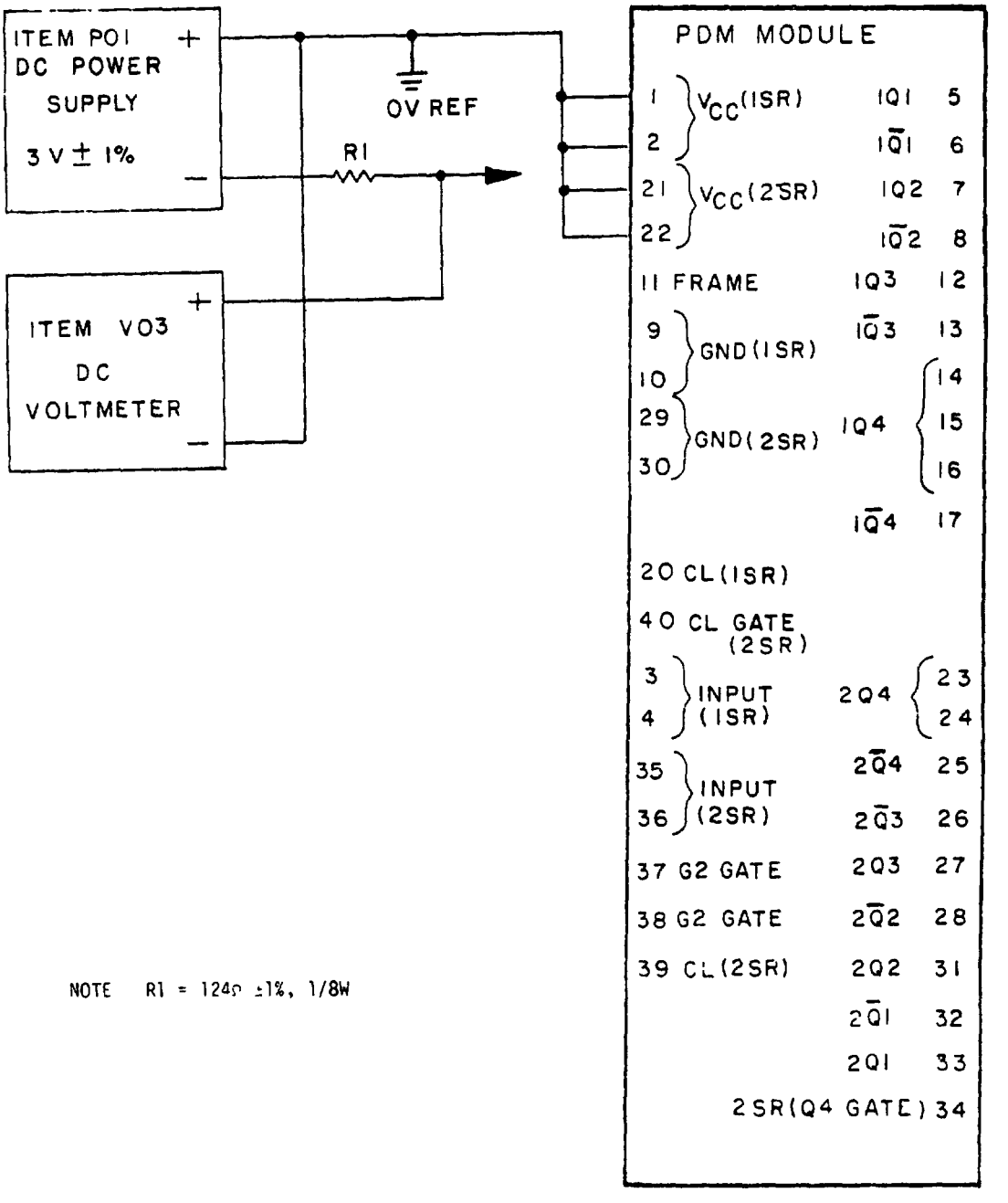
FIGURE 6 Input current test circuit for table IV.

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TABLE V. Input clamp diode voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	1SR Input clamp diode voltage	3,4	-0.4	-1.5	V	7
C2	2SR	35,36	↓	↓	↓	↓
C3	2SR GATE Q4	37				
C4	2SR GATE Q4	38				
C5	2SR CL	39				
C6	1SR CL	20				

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NOTE R1 = 124Ω ±1%, 1/8W

FIGURE 7 Input clamp diode voltage test circuit for table V

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TABLE VI. Input current (high-level and low-level) and output short circuit current test.

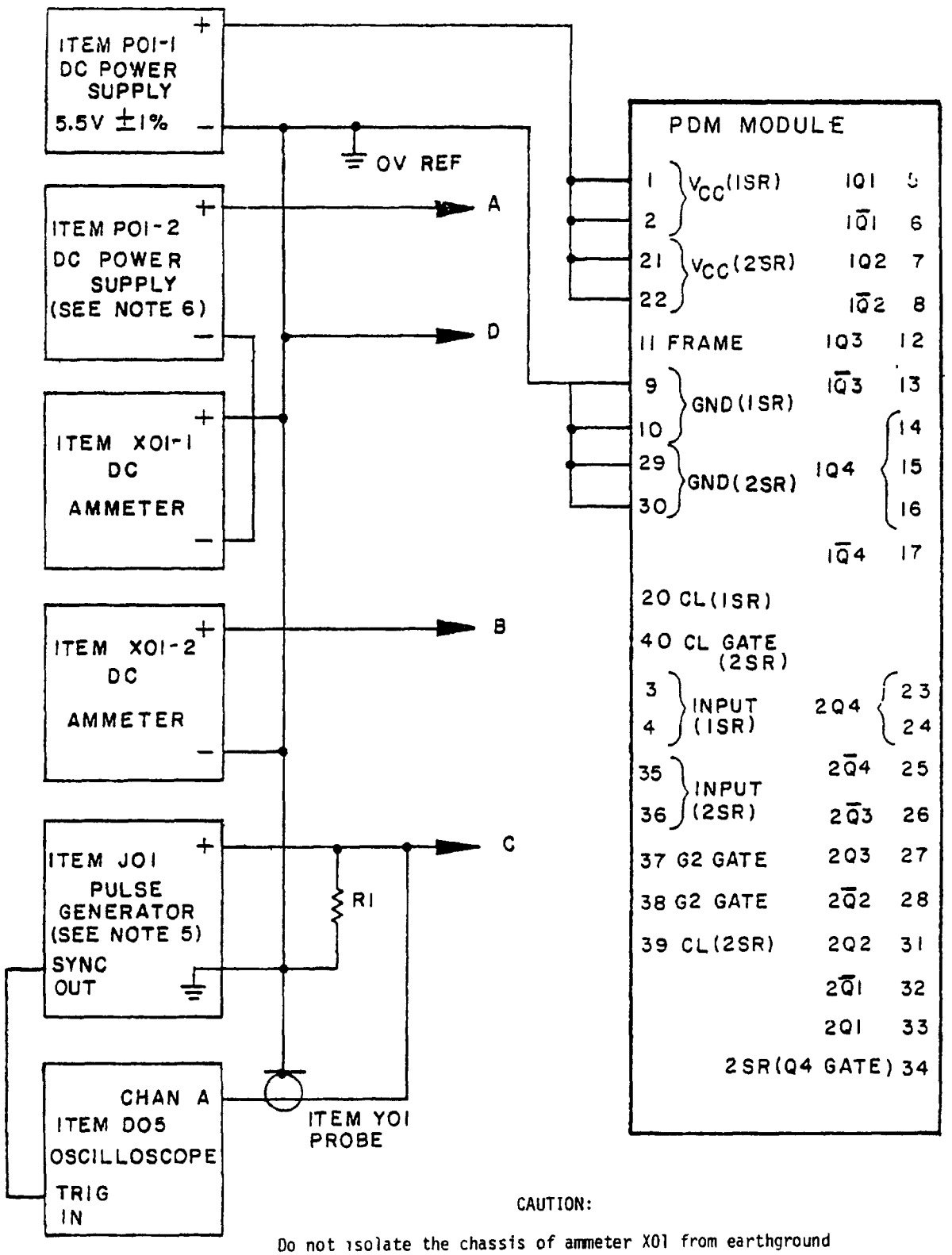
Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
D1	1SR INPUT	Input current, high-level	3,4	Open	Open	20	0	60	µA	8
D2	2SR INPUT		35,36			Open				
D3	2SR GATE Q4		37			38				
D4	2SR GATE CL		38			37				
D5	2SR CL		39			Open				
D6	1SR CL		20			3,4		360		
E1	1SR CL	Input current, low level	20	3,4		Open	0.10	1.73	mA	
E2	2SR INPUT		Open	35,36				1.73		
E3	2SR GATE Q4		38	37				2.20		
E4	2SR GATE Q4		37	38				2.20		
E5	2SR CL		Open	39				2.20		
E6	1SR INPUT		3.4	20				10.40		
F1	1Q1	Output short circuit current, high level		5	39		15	80		
F2	1Q2			7						
F3	1Q3			12						
F4	1Q4			14,15,16						

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TABLE VI. Input current (high-level and low level) and output short circuit current test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F5	201 Output short circuit current, high level	34,36	33	39	Open	15	80	mA	8
F6	2Q2		31						
F7	2Q3		27						
F8	2Q4		23,24						
F9	1$\overline{Q}$1	Open	6		3,4				
F10	1$\overline{Q}$2		8						
F11	1$\overline{Q}$3		13						
F12	1$\overline{Q}$4		17						
F13	2$\overline{Q}$1		32		35,36				
F14	2$\overline{Q}$2		28						
F15	2$\overline{Q}$3		26						
F16	2$\overline{Q}$4		25						
F17	2SR GATE Q4		34	Open	37,38	40	100		
F18	2SR CL		40	Open	39	40	100		

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CAUTION:

Do not isolate the chassis of ammeter X01 from earthground

FIGURE 8 Input current (high-level and low-level) and output short circuit current test circuit for table VI

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NOTES
1 R1 = 50 Ω 5%, 1W
2 Since ammeters X01-1 and X01-2 are not used simultaneously, only one ammeter X01 is required
3 With point A disconnected, ammeter X01-1 should indicate 0.0mA on the 1μA range
4 After point C has been connected to the appropriate pin and the module clock pulsed (none for test conditions D1-D6, E1-E6, F17 and F18, 8 pulses for test conditions F1, F5, F9 and F13, 16 pulses for test conditions F2, F6, F10, and F14, 24 pulses for test conditions F3, F7, F11 and F15, 32 pulses for test conditions F4, F8, F12 and F16), the short circuit measurement can be made
5 Item J01 setup
  PA = 4 0 ±0 5V
  PW = 1 0 ±0 1μs.
  $t_{TLH} = t_{THL} = 25 \pm 5ns$
6 Adjust item P01-2 for 4 5V ±1% for test conditions D and F, 2 4V ±1% for test condition E.
7 For test conditions F1-F16, connect pin 20 to pin 40

FIGURE 8. Input current (high-level and low-level) and output short circuit current test circuit for table VI - Continued

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TABLE VII. Output voltage (high-level and low-level) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
G1	1Q1 Output voltage, high level (input threshold high)	3,4	5	Open	2.4	4.5	V	9
G2	1Q2		7					
G3	1Q3		12					
G4	1Q4		14,15,16					
G5	2Q1	35,36	33					
G6	2Q2		31					
G7	2Q3		27					
G8	2Q4		23,24					
G9	1$\overline{Q}$1 Output voltage, high level (input threshold low)	3,4	6					
G10	1$\overline{Q}$2		8					
G11	1$\overline{Q}$3		13					
G12	1$\overline{Q}$4		17					
G13	2$\overline{Q}$1	35,36	32					
G14	2$\overline{Q}$2		28					
G15	2$\overline{Q}$3		26					
G16	2$\overline{Q}$4		25					
G17	2SR GATE		34	37,38				
G18	2SR GATE	38	34	35,36,37				
G19	2SR GATE	37	34	35,36,38				
G20	1SR GATE	39	20,40	Open				

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TABLE VII. Output voltage (high-level and low-level) test - Continued

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1Q1 Output voltage, low level (input threshold low)	3,4	5	Open	0.00	0.40	V	9
H2	1Q2		7					
H3	1Q3		12					
H4	1Q4		14,15,16					
H5	2Q1	35,36	33					
H6	2Q2		31					
H7	2Q3		27					
H8	2Q4		23,24					
H9	$1\overline{Q}1$ Output voltage, low level (input threshold high)	3,4	6					
H10	$1\overline{Q}2$		8					
H11	$1\overline{Q}3$		13					
H12	$1\overline{Q}4$		17					
H13	$2\overline{Q}1$	35,36	32					
H14	$2\overline{Q}2$		28					
H15	$2\overline{Q}3$		26					
H16	$2\overline{Q}4$		25					
H17	2SR GATE		34	37,38				
H18	2SR GATE	38	34	35,36,37				
H19	2SR GATE	37	34	35,36,38				
H20	1SR GATE	39	20,40	Open				

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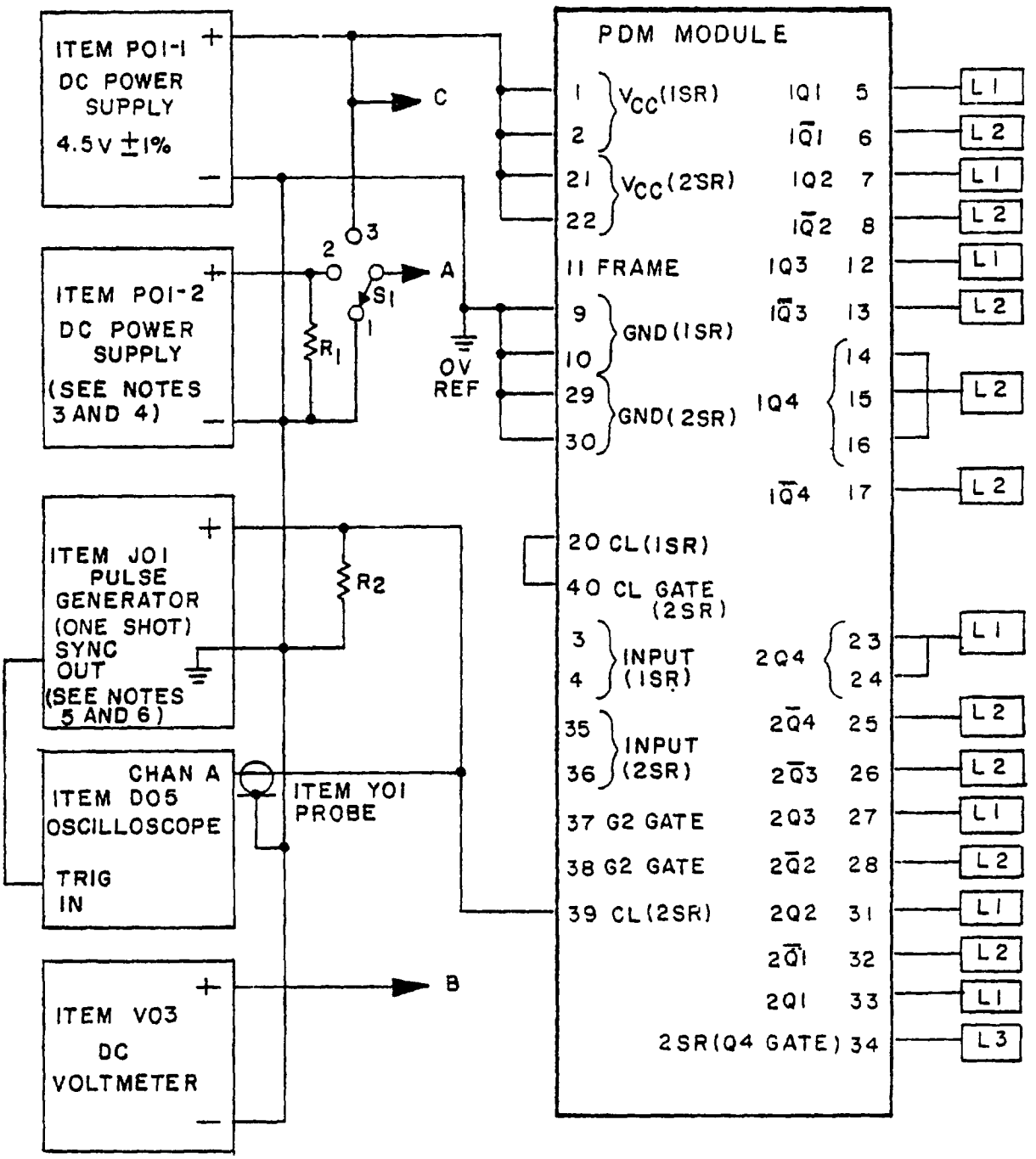


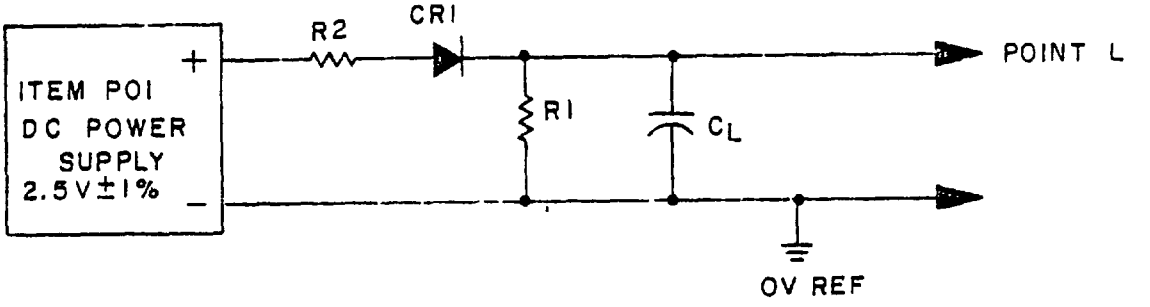
FIGURE 9 Output voltage (high-level and low-level) test circuit for table VII

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NOTES:
1 R1 = R2 = 50Ω ±5%, 1W
2 Load outputs with load circuits (L1, L2, L3) as specified on figure 10
3 For test conditions G1-G8 and H9-H20 set item P01-2 as follows.
  At 0° - 2.0V ±1%.
  25° - 1.9V ±1%
  80° - 1.8V ±1%
4 For test condition G9-G20 and H1-H8 set item P01-2 as follows
  At 0° - 0.8V ±1%
  25° - 0.9V ±1%.
  80° - 0.8V ±1%.
5 Switching sequence
  a For test conditions G1-G8 and H9-H16
    1 Place switch S1 (single pole, 3 position) in position 1
    2 After the module clock has been pulsed (8 pulses for test conditions G1, G5, H9 and H13, 16 pulses for test conditions G2, G6, H10 and H14, 24 pulses for G3, G7, H11 and H15, and 32 pulses for test conditions G4, G8, H12 and H16) place switch S1 in position 2.
    3. Pulse the module again corresponding to the exact number of clock pulses specified in 5 a 2 and make measurement
  b For test conditions G9-G16 and H1-H8
    1 Place switch S1 in position 3.
    2 Repeat 5 a 2 and 5.a 3 (8 pulses for test conditions G9, G13, H1 and H5, 16 pulses for test conditions G10, G13, H2 and H6, 24 pulses for test conditions G11, G15, H3 and H7. 32 pulses for test conditions G12, G16, H4 and H8).
  c For test conditions G17-G19 and H17-H19.
    1 Place switch S1 in position 2
    2 Perform 5 a 3 (32 pulses for test conditions G17-G19 and H17-H19)
6 Item J01 setup
  PA = 4.0 ±0.5V
  PW = 1.0 ±0.1µs.
  $t_{TLH} = t_{THL} = 25 \pm 5ns$

FIGURE 9 Output voltage (high-level and low-level) test circuit for table VII - Continued

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NOTES
1. Load circuit 1
   CR1 = 1N4150 or equivalent.
   C_L = 94 ± 5 pF (including probe and parasitic capacitance)
   R1 = 8.06 kΩ ±1%, 1/8W.
   R2 = Select value for a current flow of 8.7 mA from point L to ground
2. Load circuit 2.
   CR1 = 1N4150 or equivalent.
   C_L = 94 ±5 pF (including probe and parasitic capacitance).
   R1 = 6.65 kΩ ±1%, 1/8W.
   R2 = Select value for a current of 10.4 mA from point L to ground
3. Load circuit 3.
   CR1 = 1N4150 or equivalent.
   C_L = 130 ±5 pF (including probe and parasitic capacitance)
   R1 = 1.91 kΩ ±1%, 1/8W.
   R2 = Select value for a current flow of 22 mA from point L to ground
4. Capacitor C_L is not required for dc electrical test conditions G and H.
5. During output switching, the 2.5V power supply shall have a maximum overshoot of 500 mV peak.

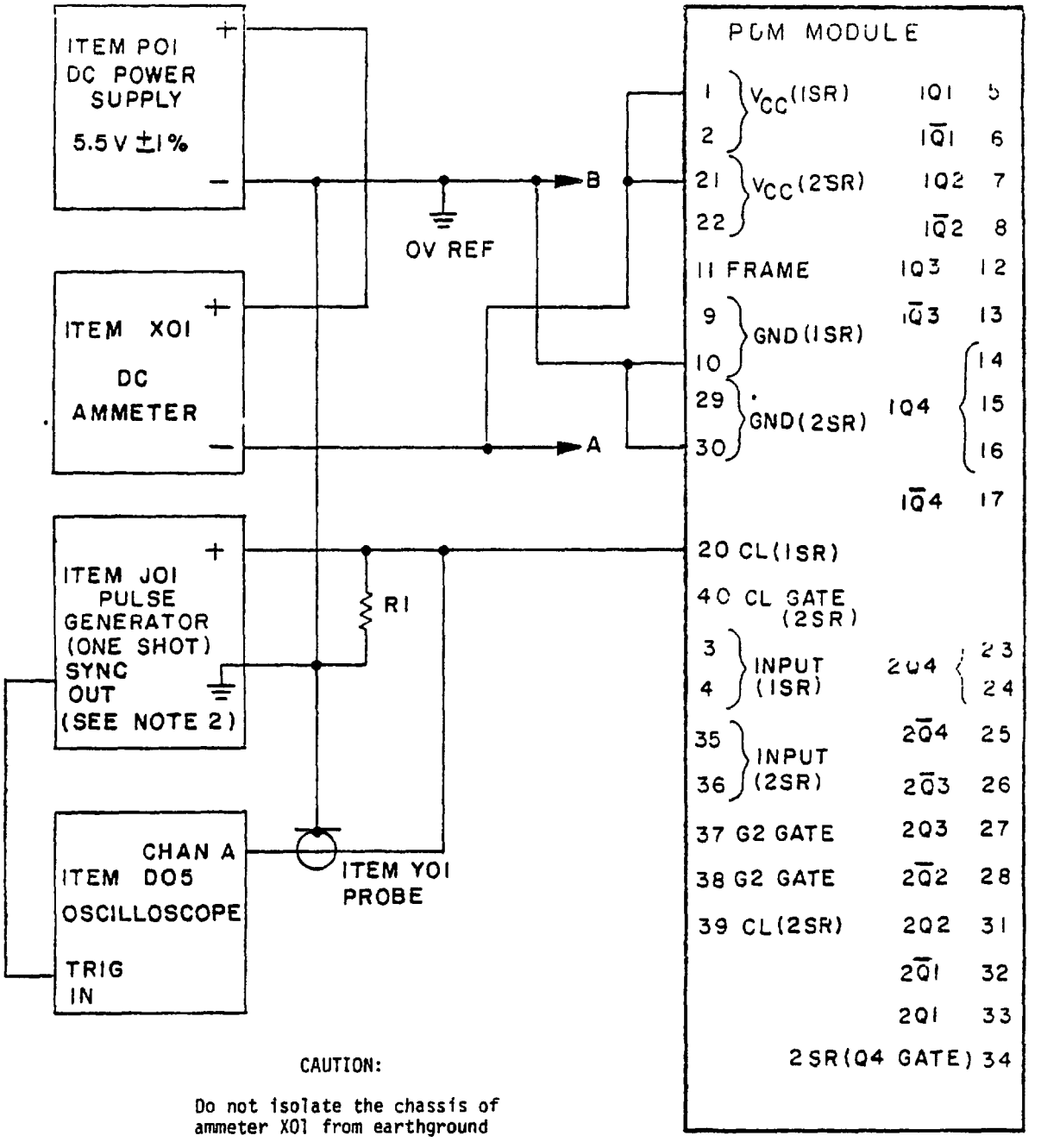
FIGURE 10. PDM logic load circuit.

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TABLE VIII Power supply current (high-level and low-level) test

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Power supply current, high level	3,4,35,36	Open	100	345	mA	11
J2	Power supply current, low level	Open	3,4,35,36	100	345	mA	11

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CAUTION:

Do not isolate the chassis of ammeter X01 from earthground

NOTES.
1 R1 = 50Ω ±5%, 1W
2 Pulse pin 3, 4, 35, 36 for 39 pulses before the measurements are taken Item J01 setup.
PA = 4.0 ±0.5V
PW = 1.0 ±0.1µs
$t_{TLH} = t_{THL}$ = 25 ±5ns.

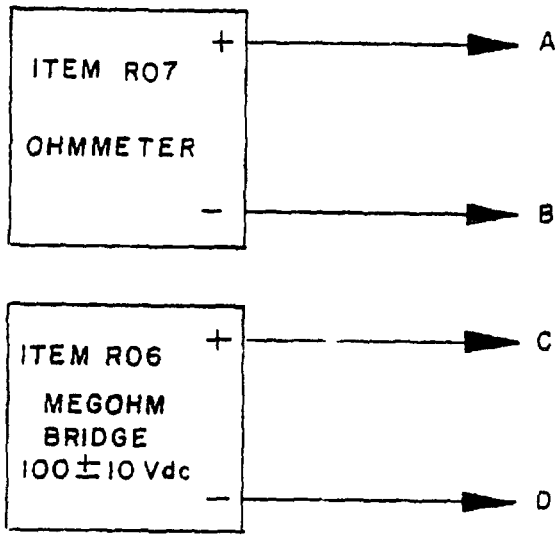
FIGURE 11. Power supply current (high-level and low-level) test circuit for table VIII.

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TABLE IX. Pin continuity and isolation test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
K1	1NPUT 1SR Pin continuity	3	4	Open	Open	0	1	Ω	12
K2	1Q4	14	15						
K3	1Q4	14	16						
K4	1Q4	15	16						
K5	INPUT 2SR	35	36						
K6	2Q4	23	24						
K7	1SR	1	2						
K8	1SR	9	10						
K9	2SR	21	22						
K10	2SR	29	30						
K11		11	Frame						
L	Isolation	Open	Open	All other pins	11	10	∞	MΩ	

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WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition K11, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

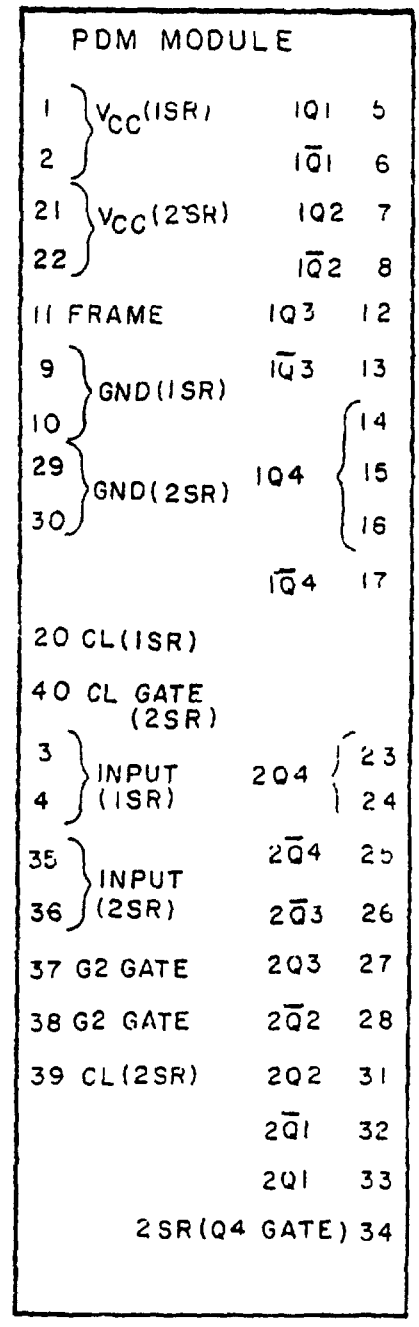


FIGURE 12. Pin continuity and isolation test circuit for table IX.

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TABLE X. Transition time (t_{TLH}) test

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
M1	1SR GATE	Transition time (t_{TLH})	Open	39	20,40	2	20	ns	10,11
M2	2RS GATE		Open	37	34	2			
M3	1Q1		3,4	39	5	5			
M4	1Q2				7				
M5	1Q3				12				
M6	1Q4				14,15,16				
M7	1$\overline{Q}$1				6				
M8	1$\overline{Q}$2				8				
M9	1$\overline{Q}$3				13				
M10	1$\overline{Q}$4				17				
M11	2Q1		35,36		33				
M12	2Q2				31				
M13	2Q3				27				
M14	2Q4				23,24				
M15	2$\overline{Q}$1				32				
M16	2$\overline{Q}$2				28				
M17	2$\overline{Q}$3				26				
M18	2$\overline{Q}$4				25				

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TABLE XI. Transition time (t_{THL}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
N1	1SR GATE Transition time (t_{THL})	Open	39	20,40	2	20	ns	13,14
N2	2RS GATE	Open	37	34	2			
N3	1Q1	3,4	39	5	5			
N4	1Q2			7				
N5	1Q3			12				
N6	1Q4			14,15,16				
N7	1$\overline{Q}$1			6				
N8	1$\overline{Q}$2			8				
N9	1$\overline{Q}$3			13				
N10	1$\overline{Q}$4			17				
N11	2Q1	35,36		33				
N12	2Q2			31				
N13	2Q3			27				
N14	2Q4			23,24				
N15	2$\overline{Q}$1			32				
N16	2$\overline{Q}$2			28				
N17	2$\overline{Q}$3			26				
N18	2$\overline{Q}$4			25				

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TABLE XII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P1	1SR GATE	Propagation delay time (t_{PLH})	Open	39	20,40	2	25	ns	13,14
P2	2RS GATE		Open	37	34	2	25		
P3	1Q1		3,4	39	5	5	57		
P4	1Q2				7				
P5	1Q3				12				
P6	104				14,15,16				
P7	$1\overline{Q}1$				6				
P8	$1\overline{Q}2$				8				
P9	$1\overline{Q}3$				13				
P10	$1\overline{Q}4$				17				
P11	2Q1		35,36		33				
P12	2Q2				31				
P13	2Q3				27				
P14	2Q4				23,24				
P15	$2\overline{Q}1$				32				
P16	$2\overline{Q}2$				28				
P17	$2\overline{Q}3$				26				
P18	$2\overline{Q}4$				25				

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TABLE XIII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter		Connect			Initial limits 25 ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
01	1SR GATE	Propagation delay time (t_{PHL})	Open	39	20,40	2	25	ns	13,14
02	2RS GATE		Open	37	34	2	25		
03	1Q1		3,4	39	5	5	73		
Q4	1Q2				7				
Q5	1Q3				12				
Q6	1Q4				14,15,16				
Q7	1$\overline{Q}$1				6				
Q8	1$\overline{Q}$2				8				
09	1$\overline{Q}$3				13				
Q10	1$\overline{Q}$4				17				
011	2Q1		35,36		33				
012	2Q2				31				
013	2Q3				27				
014	2Q4				23,24				
015	2$\overline{Q}$1				32				
016	2$\overline{Q}$2				28				
017	2$\overline{Q}$3				26				
018	2$\overline{Q}$4				25				

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TABLE XIV. Maximum clock frequency and minimum set-up and hold time check test

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R1	1SR	Maximum clock frequency and minimum set-up time check	3,4	20	14,15,16	25	98	ns	13,15
R2	1SR	↓	3,4	20	17	↓	98	↓	13,15
R3	2SR	↓	35,36	39	23,24	↓	121	↓	13,16
R4	2SR	↓	35,36	39	25	↓	121	↓	13,16
R5	1SR	Maximum clock frequency and minimum hold time check	3,4	20	14,15,16	5	73	↓	13,15
R6	1SR	↓	3,4	20	17	5	73	↓	13,15
R7	2SR	↓	35,36	39	23,24	7	98	↓	13,16
R8	2SR	↓	35,36	39	25	7	98	↓	13,16

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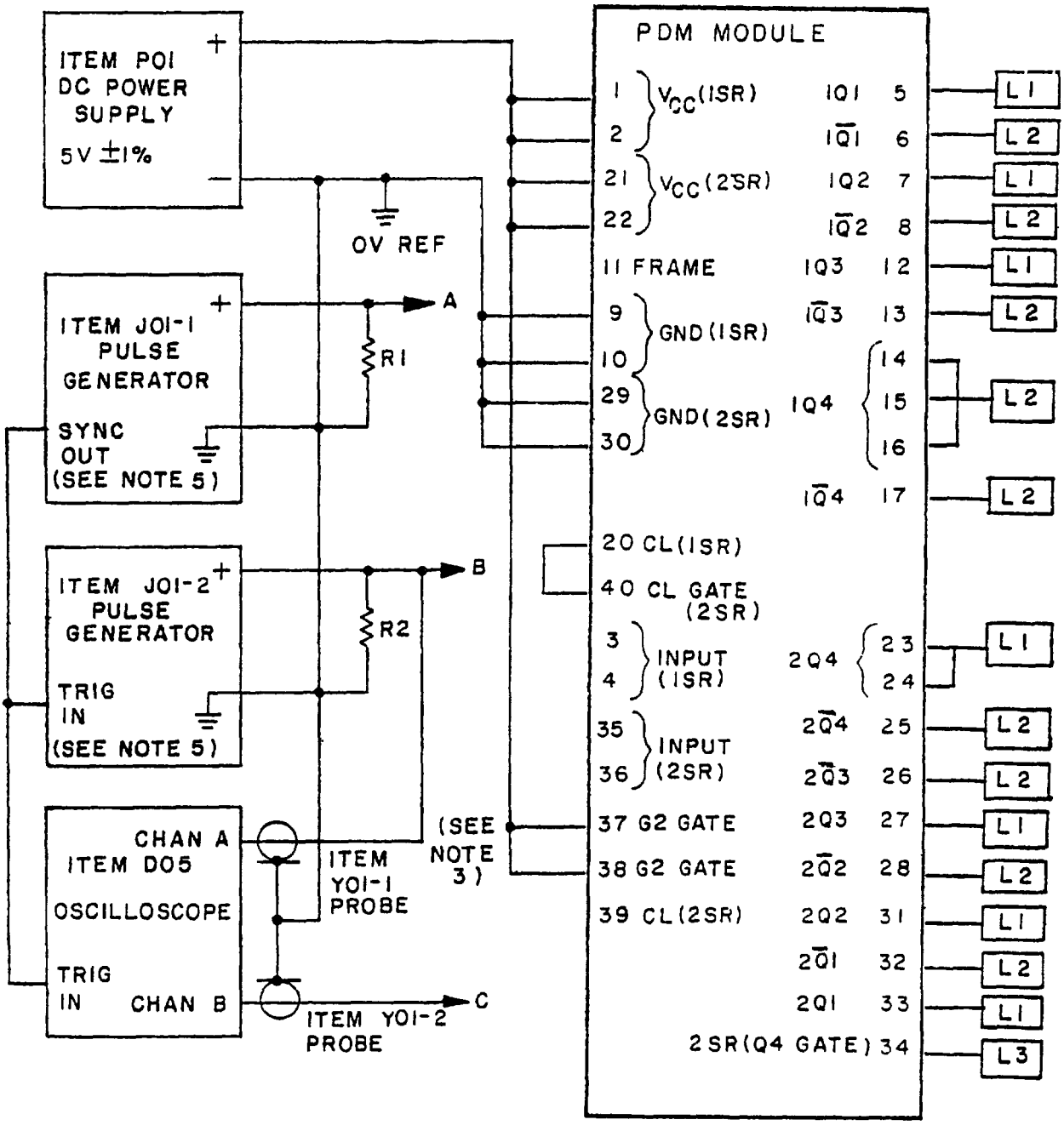


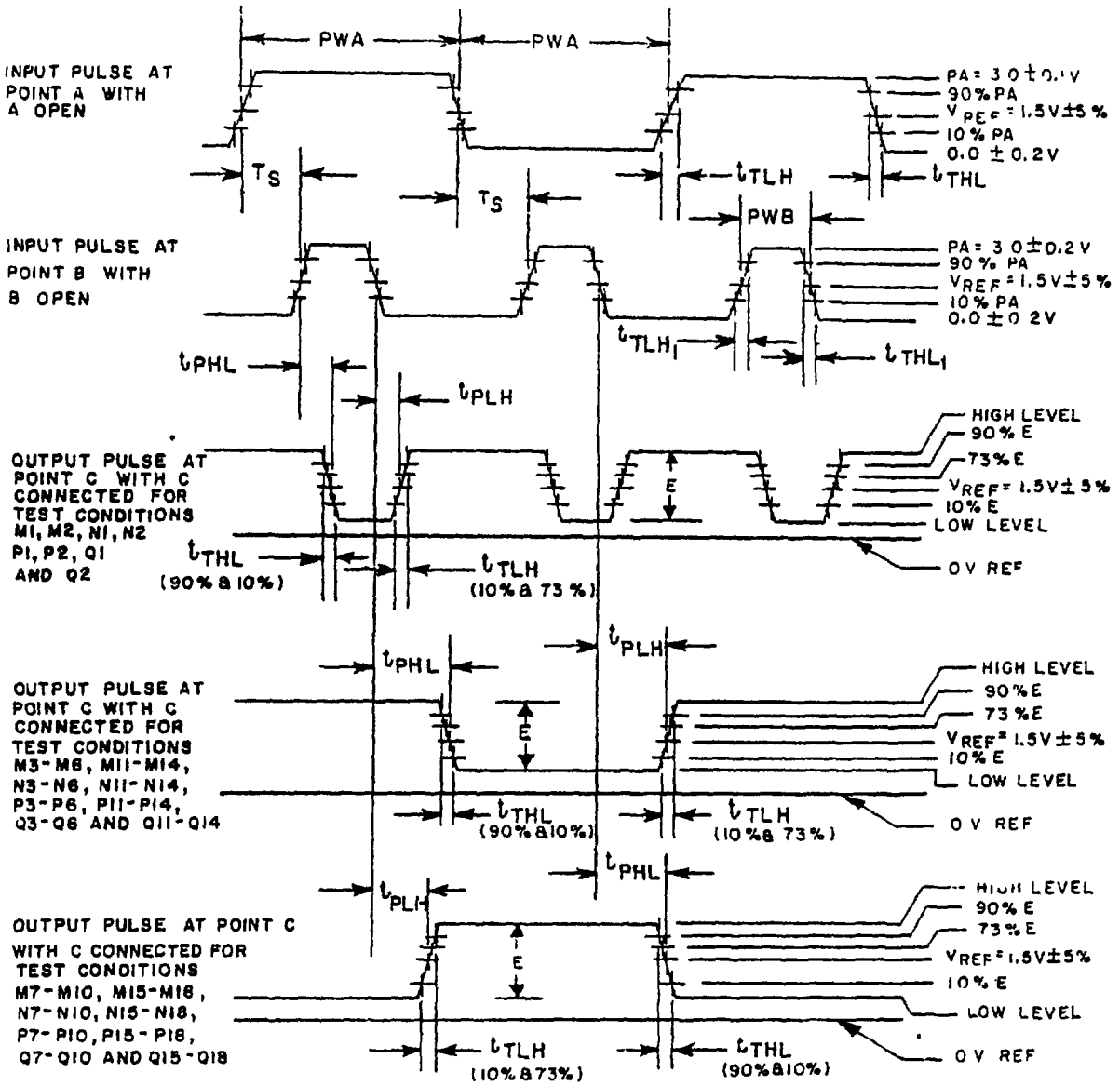
FIGURE 13. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PLH} and t_{PHL}) and maximum clock frequency and minimum setup and hold time check test circuit for tables X through XIV.

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NOTES
1. P1 = P2 = 50 Ω 5%, 1W
2. Load outputs with load circuits (L1, L2, L3) as specified on figure 10.
3. For test conditions M2, N2, P2 and Q2, initially set 2SR (GATE Q4) output in the high level state (pin 39 shall be pulsed a minimum of 32 times, with pins 35 and 36 connected to pin 1) During test conditions other than these, pin 37 shall be connected to pin 1.
4. For test conditions M, N, P, Q and R, connect point B to the appropriate module pin for a minimum of 32 clock pulses before a measurement is performed
5. Item J01-1 and J01-2 set-up procedure.
   a. Adjust item J01-1 for a waveform at point A per figure 14
   b. Place item J01-2 in the (-) ext trig repetition rate and single pulse mode Adjust for a pulse as specified at point B per figure 14 (During a complete period of waveform A, only one pulse will appear in waveform B).
   c. Place item J01-2 in the double pulse mode Adjust item J01-1 and J01-2 delays for the delays specified between waveforms of point A and B (If waveform A is not as specified, the delays between waveforms A and B cannot be obtained).

FIGURE 13 Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PLH} and t_{PHL}) and maximum clock frequency and minimum setup and hold time check test circuit for tables X through XIV - Continued.

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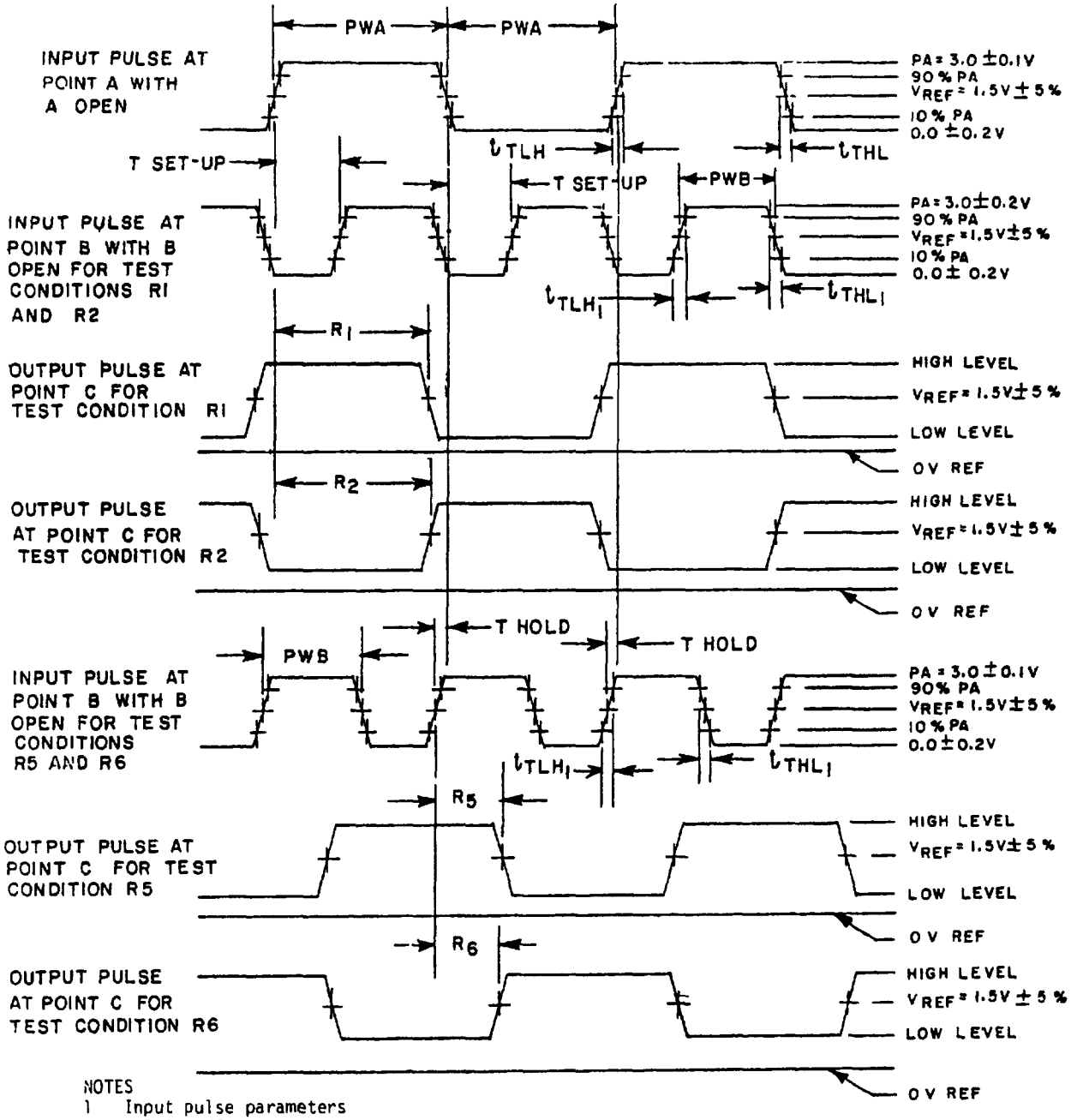
NOTES:

1 Input pulse parameters
PWA = 300 ±30 ns.
PWB = 100 ±10 ns
$t_{TLH} = t_{THL}$ = 25 ±5 ns.
$t_{TLH_1} = t_{THL_1}$ = 6 ±2 ns.
t_s = 100 ±10 ns.

2 Verify that all outputs have a high level ≥ 2.4V and a low level ≤ 0.4V before making the measurements.

FIGURE 14 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables X through XIII.

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NOTES

1 Input pulse parameters

PWA = 62 ±1 ns

PWB = 35 ±1 ns

$t_{TLH} = t_{THL} = 25 \pm 5$ ns.

$t_{TLH_1} = t_{THL_1} = 6 +2$ ns

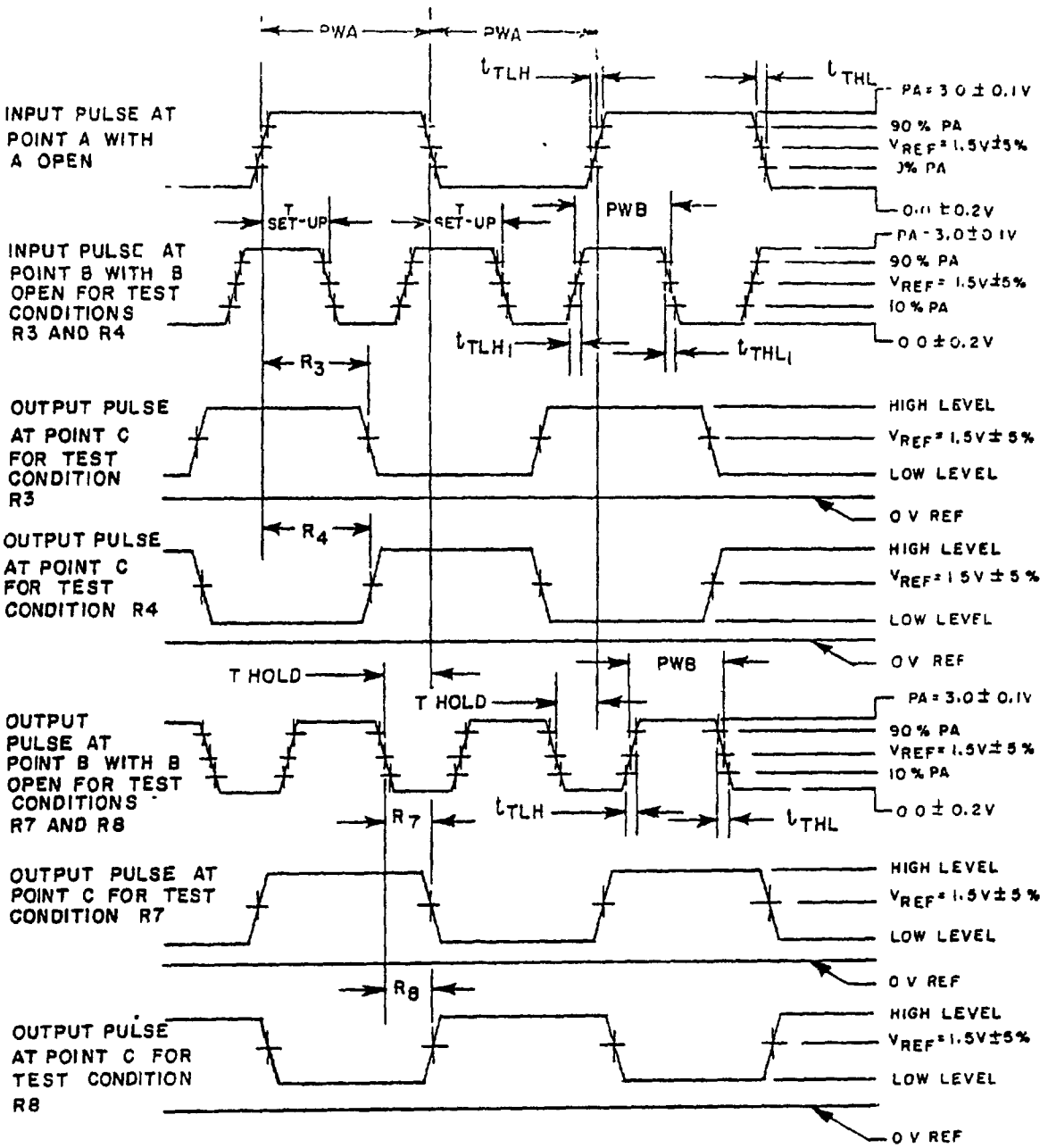
$T_{SET-UP} = 25 \pm 1$ ns

$T_{HOLD} = 0 +1$ ns

2 Verify that all outputs have a high level ≥ 2 4V and a low level 0 4V before making the measurements

FIGURE 15 Maximum clock frequency and minimum set-up and hold time check test circuit waveforms for table XIV

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NOTES.

1. Input pulse parameters:
   PWA = 62 ±1 ns
   PWB = 27 ±1 ns.
   $t_{TLH} = t_{THL}$ = 25 ±5 ns.
   $t_{TLH_1} = t_{THL_1}$ = 6 ±2 ns
   T_{SET-UP} = 23 ±1 ns.
   T_{HOLD} = 25 +2 ns
2. Verify that all outputs have a high level ≥ 2.4 V and a low level ≤ 0.4V before making the measurements.

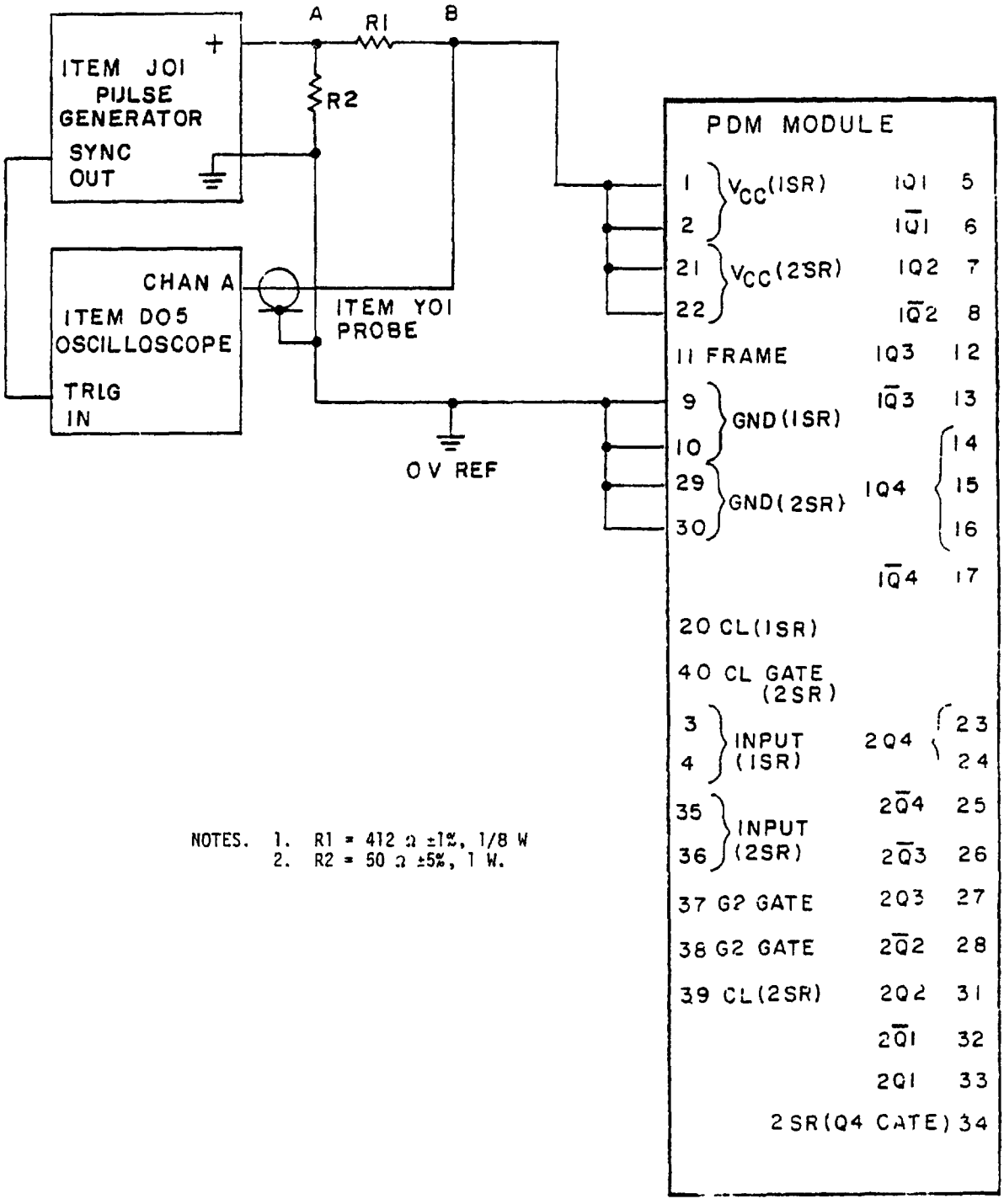
FIGURE 16 Maximum clock frequency and minimum set-up and hold time check test circuit waveforms for tabel XIV.

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TABLE XI. Power supply transient rise time test

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
c	Power supply transient rise time	2.5	7.5	µs	4, 18

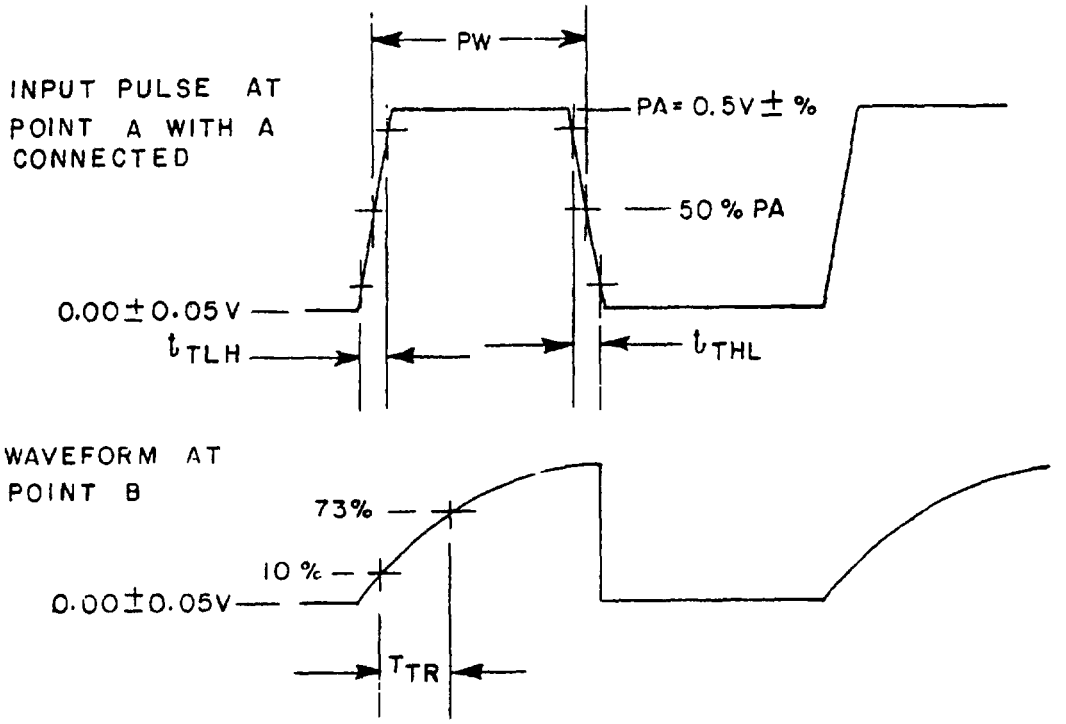
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NOTES. 1. R1 = 412 Ω ±1%, 1/8 W
2. R2 = 50 Ω ±5%, 1 W.

FIGURE 17. Power supply transient rise time test circuit for table XV.

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NOTES 1. Input pulse parameters
PW = 30 µs ±10%.
PRP = 60 µs ±10%.
$t_{TLH} = t_{THL}$ = 50 ns ±10%.

2. Measure transient rise time (t_{TR}) at point B.

FIGURE 18. Power supply transient rise time test circuit wave forms for table XV.

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TABLE XVI. Output crosstalk (high-level and low-level) test.

Test condition	Parameter		Connect							Initial limits 25°C±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
T1	101	Output crosstalk, high level	3,4	35,36	39	5	Open	5	Open	0.0	0.4	V	19,20
T2	102					7	Open	7					
T3	103					12	Open	12					
T4	104					14,15,16	14,15,16	Open					
T5	$10\bar{1}$					6	6						
T6	$10\bar{2}$					8	8						
T7	$10\bar{3}$					13	13						
T8	$10\bar{4}$					17	17						
T9	201		35,36	3,4	20	33	Open	33					
T10	202					31		31					
T11	203					27		27					
T12	204					23,24		23,24					
T13	$20\bar{1}$					32	32	Open					
T14	$20\bar{2}$					28	28						
T15	$20\bar{3}$					26	26						
T16	$20\bar{4}$					25	25						
T17	ISP CL		39			40	40						
T18	ISR (GATE 04)		35,36			34	Open		34				

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TABLE VI Output crosstalk (high-level and low-level) test Continued.

Test condition	Parameter	Connect A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Initial limits 25°C±5°C ambient Min	Max	Unit	Test figure
U1	1Q1 Output crosstalk, high level	3,4	35,36	39	5	Open	5	Open	0.0	0.4	V	10,20
U2	1Q2	↓	↓	↓	7	Open	7	↓	↓	↓	↓	↓
U3	1Q3	↓	↓	↓	12	Open	12	↓	↓	↓	↓	↓
U4	1Q4	↓	↓	↓	14,15,16	14,15,16	Open	↓	↓	↓	↓	↓
U5	1$\overline{Q}$1	↓	↓	↓	6	6	↓	↓	↓	↓	↓	↓
U6	1$\overline{Q}$2	↓	↓	↓	8	8	↓	↓	↓	↓	↓	↓
U7	1$\overline{Q}$3	↓	↓	↓	13	13	↓	↓	↓	↓	↓	↓
U8	1$\overline{Q}$4	↓	↓	↓	17	17	↓	↓	↓	↓	↓	↓
U9	2Q1	35,36	3,4	20	33	Open	33	↓	↓	↓	↓	↓
U10	2Q2	↓	↓	↓	31	↓	31	↓	↓	↓	↓	↓
U11	2Q3	↓	↓	↓	27	↓	27	↓	↓	↓	↓	↓
U12	2Q4	↓	↓	↓	23,24	↓	23,24	↓	↓	↓	↓	↓
U13	2$\overline{Q}$1	↓	↓	↓	32	32	Open	↓	↓	↓	↓	↓
U14	2$\overline{Q}$2	↓	↓	↓	28	28	↓	↓	↓	↓	↓	↓
U15	2$\overline{Q}$3	↓	↓	↓	26	26	↓	↓	↓	↓	↓	↓
U16	2$\overline{Q}$4	↓	↓	↓	25	25	↓	↓	↓	↓	↓	↓
U17	1SR CL	39	↓	↓	40	40	↓	↓	↓	↓	↓	↓
U18	2SR (GATE Q4)	35,36	↓	↓	34	Open	↓	34	↓	↓	↓	↓

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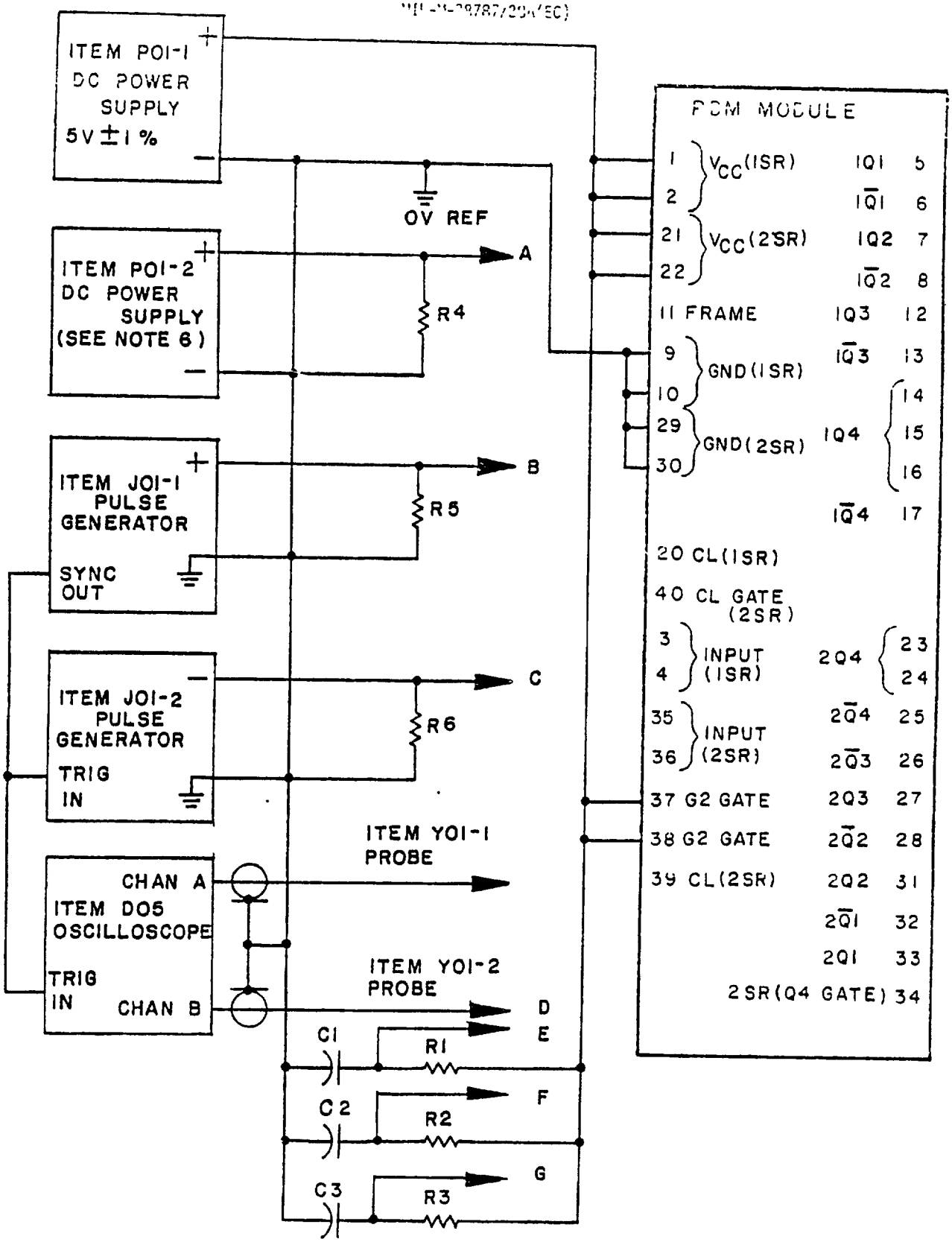


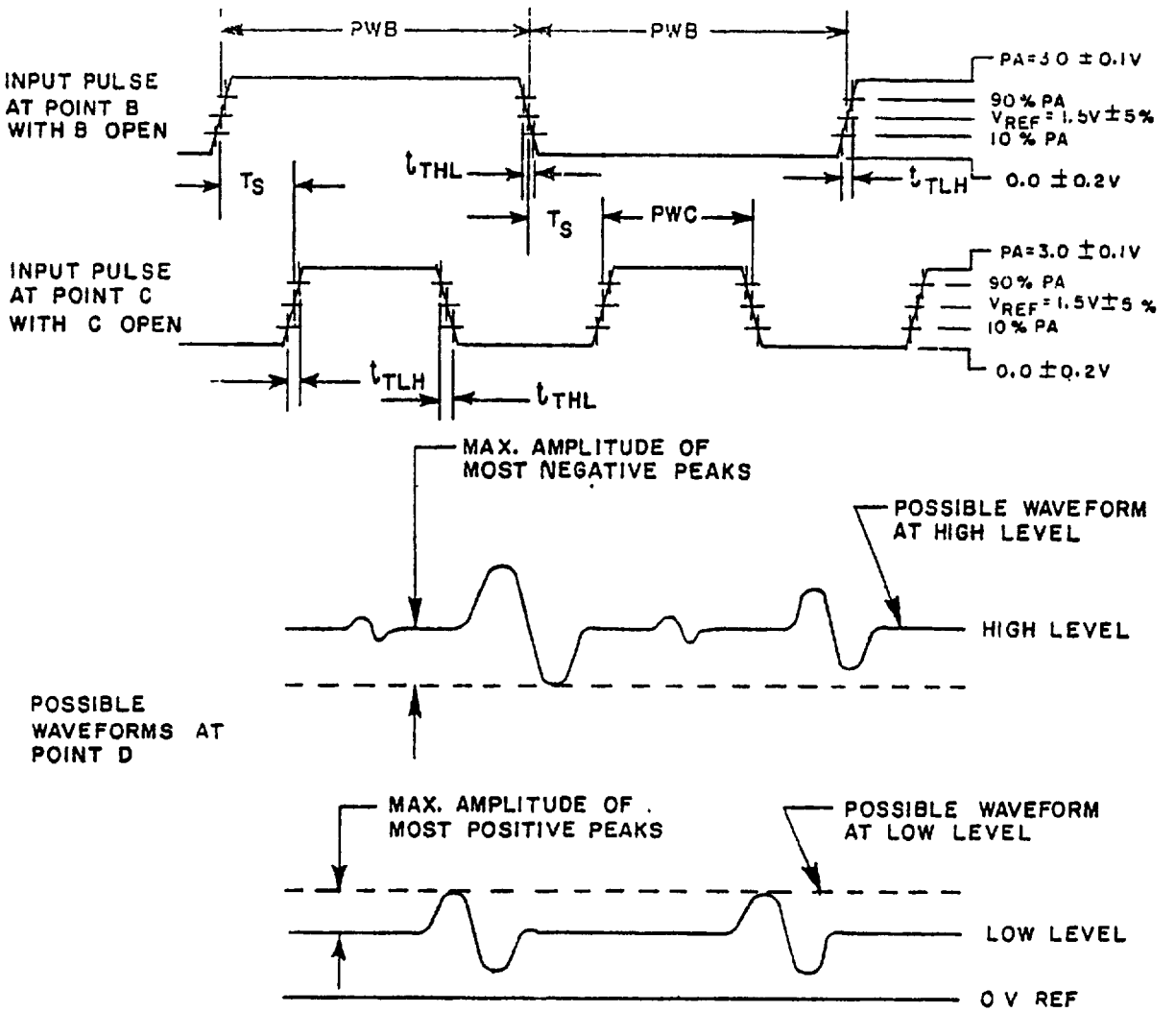
FIGURE 19. Output crosstalk (high-level and low-level) test circuit for table XVI.

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NOTES
1 R1 = 523Ω ±1%, 1/8W
2 R2 = 634Ω ±1%, 1/8W
3 R3 = 243Ω 1%, 1/4W
4 R4 = R5 = R6 = 50Ω ±5%, 1W
5 C1 = C2 = C3 = 30 ±1pF (including probe capacitance)
6 For test conditions T1-T4, T9-T12, U5-U8 and U13-U18, set item P01-2 to 2 4V ±1% and for test conditions T5-T8, T13-T18, U1-U4 and U9-U12, set item P01-2 to 0 4V ±1%
7 For test conditions T1-T8 and U1-U8, connect pin 20 to pin 40 for a minimum of 32 clock pulses Then connect pin 20 to pin 10 and make measurements (This operation will set the appropriate outputs at the proper logic level)
8 For test conditions T9-T18 and U9-U18, connect pin 39 to point C for a minimum of 32 clock pulses Then connect pin 39 to pin 1. (For test conditions T17 and U17, connect pin 39 to point A instead of pin 1)
9 The loop length from pin 1 to each resistor and output, including the length of the resistor body, shall not exceed 4.5 inches (114 3 mm).

FIGURE 19. Output crosstalk (high-level and low-level) test circuit for table XVI. - Continued

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NOTE: Input pulse parameters:
PWB = 200 ±20ns.
PWC = 100 ±10ns
T_S = 50 ±10ns
$t_{TLH} = t_{THL}$ = 6 ±2ns

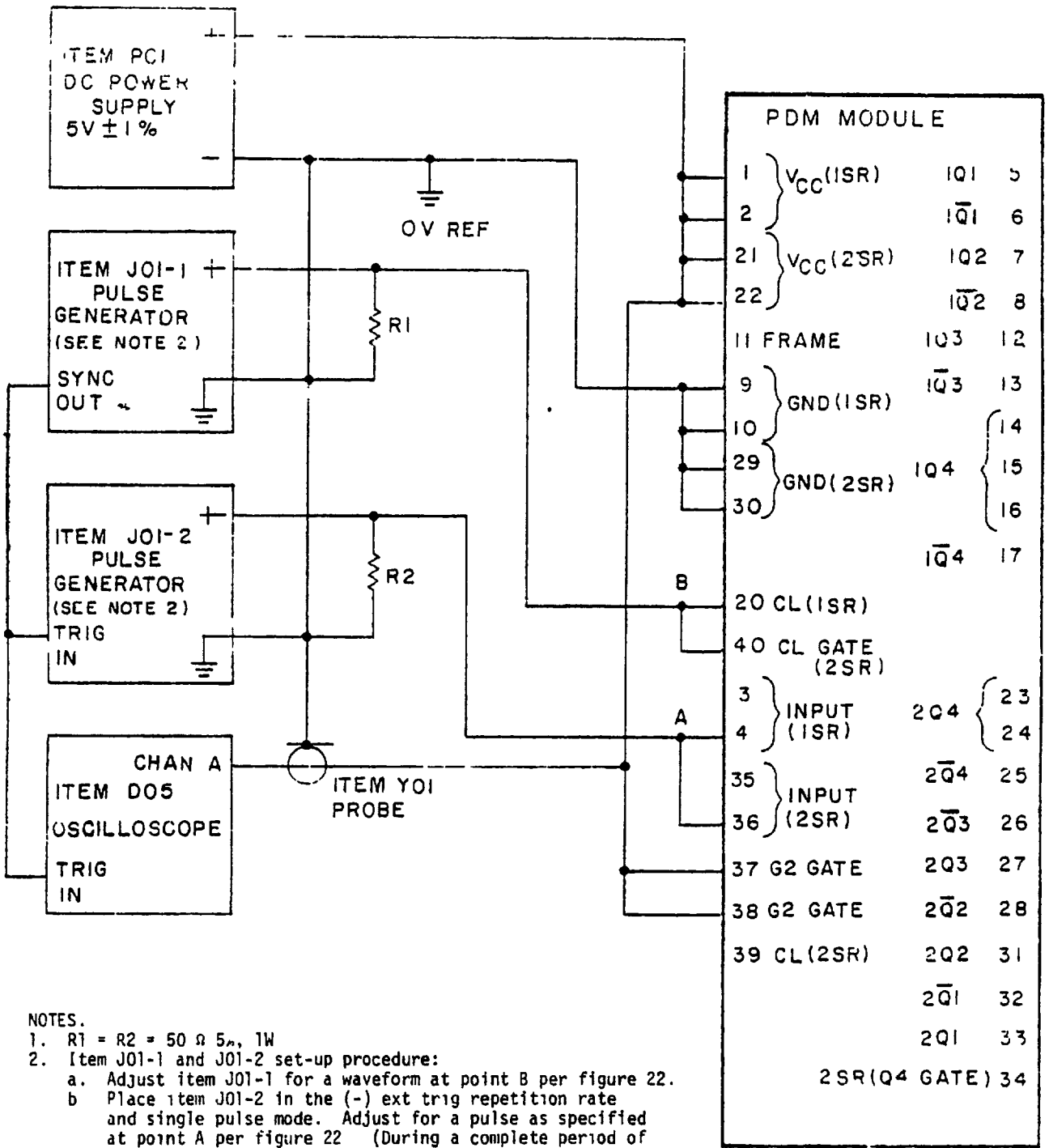
FIGURE 20 Output crosstalk (high-level and low-level) test circuit waveforms for table XVI.

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TABLE XV.I. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25 ±5°C ambient		Unit	Test figure
		Min	Max		
	Power supply transient amplitude		400		23,24

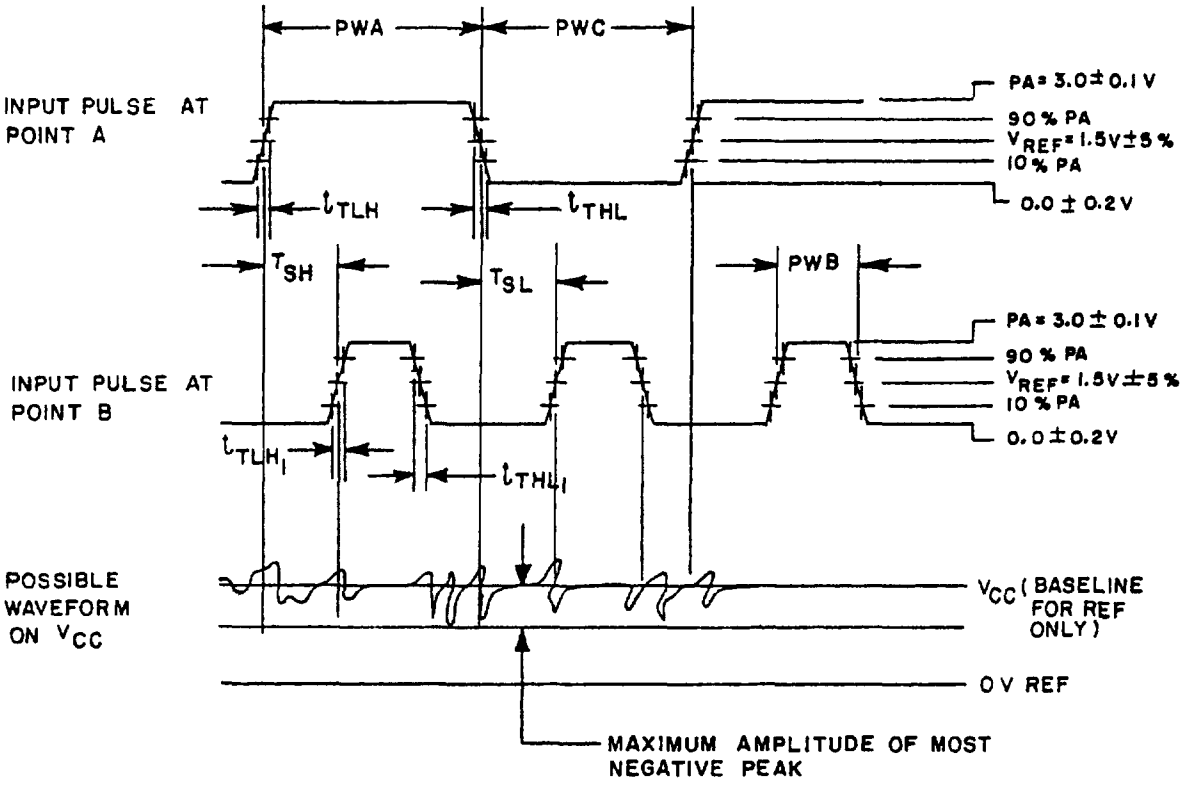
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NOTES.
1. R1 = R2 = 50 Ω 5%, 1W
2. Item J01-1 and J01-2 set-up procedure:
   a. Adjust item J01-1 for a waveform at point B per figure 22.
   b Place item J01-2 in the (-) ext trig repetition rate and single pulse mode. Adjust for a pulse as specified at point A per figure 22 (During a complete period of waveform A, only one pulse will appear in waveform B).
   c. Place item J01-2 in the double pulse mode. Adjust item J01-1 and J01-2 delays for the delays specified between waveforms on point A and B. (If waveform A is not as specified, the delays between waveforms A and B cannot be obtained)

FIGURE 21 Power supply transient amplitude test circuit for table VII

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NOTE: Input pulse parameters
PWA = 1.0 ±0.1µs
PWB = 100 ±10ns
PWC = 4.0 ±0.2µs.
T_{SH} = 450 ±50ns
T_{SL} = 2.0 ±0.1µs.
$t_{TLH} = t_{THL}$ = 25 ±5ns (verified at each input pin)
$t_{TLH_1} = t_{THL_1}$ = 6 ±2ns (verified at each input pin)

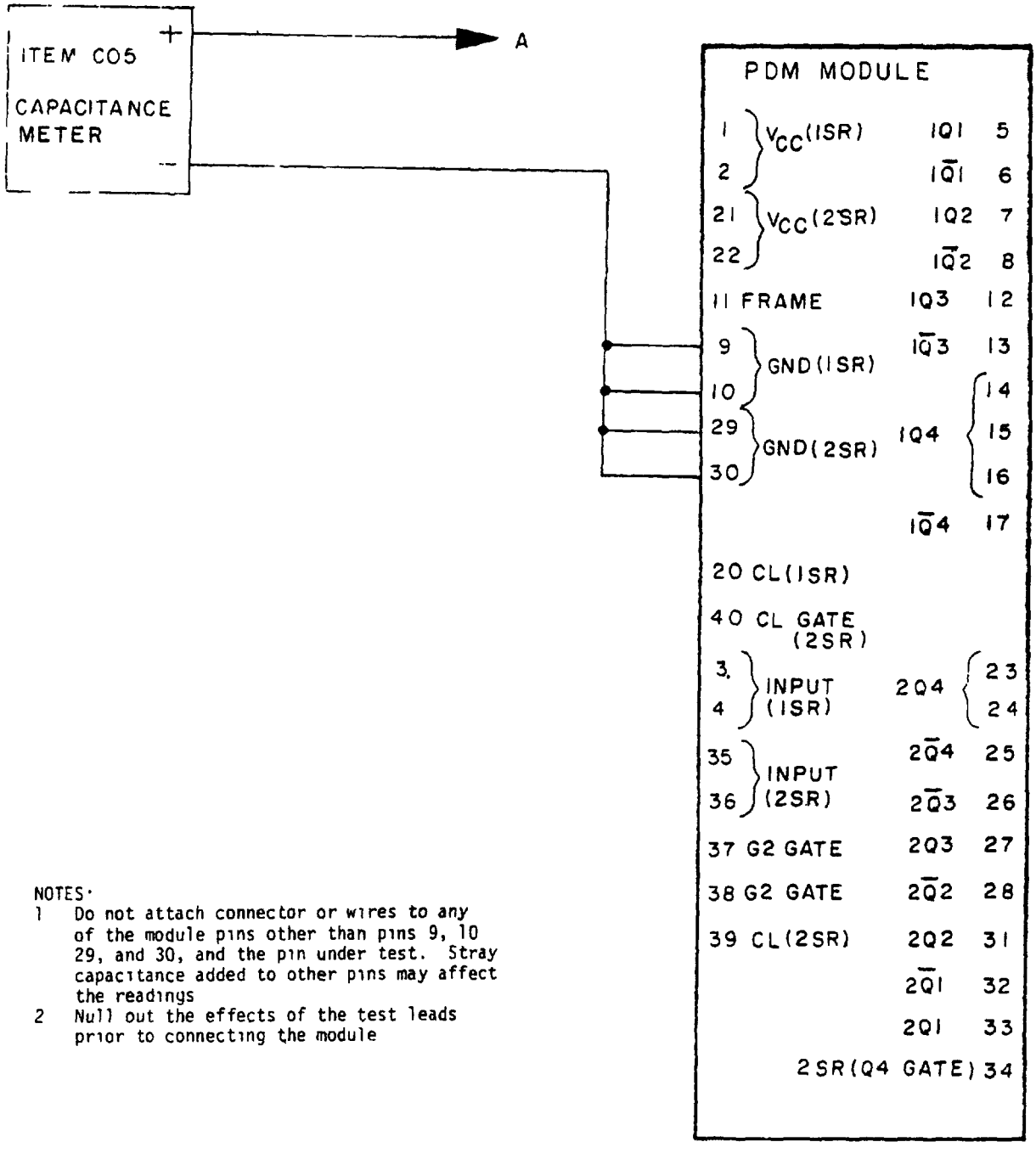
FIGURE 22. Power supply transient amplitude test circuit waveforms for table XVII

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TABLE XVIII. Input capacitance.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
W1	1SP Input capacitance	3	2	12	pF	23
W2	1SP	4				
W3	2SP	35				
W4	2SR	36				
W5	2SR GATE 4	37				
W6	2SR GATE 4	3.				
W7	2SP CL	39				
W8	1SR CL	20		48		

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NOTES:
1 Do not attach connector or wires to any of the module pins other than pins 9, 10 29, and 30, and the pin under test. Stray capacitance added to other pins may affect the readings
2 Null out the effects of the test leads prior to connecting the module

FIGURE 23 Input capacitance test circuit for table XVIII

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